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TOPICAL REVIEW

Loco-Manipulation With Quadruped Robots: Modeling, Task Taxonomy, and Control Methods–A Critical Review

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ABSTRACT Loco-manipulation–the tight, simultaneous integration of locomotion and manipulation–has rapidly expanded the autonomy and usefulness of quadruped robots equipped with arms. This survey introduces a unified modeling–tasks–controls triad that explicitly links 1) floating-base and contact modeling assumptions; 2) loco-manipulation task regimes; and 3) controller families, and uses this structure to critically evaluate the literature through evidence-based trade-off maps, failure-mode synthesis, and reported quantitative anchors enabling consistent cross-paradigm comparison beyond existing reviews. Recent advances are synthesized across *model-based* inverse-dynamics Whole-Body Control (WBC), *optimization-based* whole-body Model Predictive Control (MPC), *learning-based* Reinforcement/Imitation Learning (RL/IL), and emerging *hybrid* architectures augmented by impedance and force-control layers. Grounded in an updated survey landscape and a refined task taxonomy, the triad is used to derive "when-to-use-what" decision matrices spanning interaction type, terrain, payload, coordination, and sensing. High-fidelity simulation and digital-twin pipelines are highlighted as practical enablers for benchmarking, reproducibility, and reliable sim-to-real transfer. Using the proposed six-axis task taxonomy as the organizing lens, the hardest-to-transfer regimes consistently couple contact-rich end-effector interaction, uneven or uncertain footholds, and perception latency or occlusion. By consolidating fundamental trade-offs across control families and outlining research directions toward robust hybrid designs that couple predictive constraint handling with data-driven adaptation, this review charts a clear trajectory for deploying quadruped loco-manipulation in complex real-world environments.

INDEX TERMS Legged robot, loco-manipulation, reinforcement learning, whole-body control, optimization.

I. INTRODUCTION

Robotics is rapidly reshaping sectors such as logistics, manufacturing, and service industries by enabling higher productivity, safer operations, and greater automation in complex environments. A particularly impactful direction is *mobile manipulation*, which integrates locomotion with manipulation to allow robots to move and interact with the world in dynamic, unstructured settings [1], [2]. Within this landscape, quadruped loco–manipulation has emerged as a distinctive capability class that couples floating-base mobility with contact-rich, dexterous interaction.

Despite substantial progress in legged locomotion and manipulation, robust *simultaneous* coordination remains challenging. Many pipelines still treat locomotion and manipulation as loosely coupled modules, which can

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introduce latency, compounding tracking errors, and reduced performance in contact-rich tasks. A persistent gap in the literature is *principled guidance* that connects (i) the underlying floating-base/contact modeling assumptions, (ii) the structure of loco–manipulation tasks, and (iii) the resulting suitability of controller families (e.g., WBC, MPC, learning, and hybrids) under real-world constraints such as terrain uncertainty, payload variation, and sensing latency.

To address this gap, the present survey adopts a *modeling–tasks–controls* triad that explicitly links: (i) floating-base and contact modeling abstractions (e.g., rigid vs. compliant contacts, centroidal vs. full-body dynamics, inclusion/exclusion of object dynamics), (ii) loco–manipulation task regimes (e.g., prehensile vs. non-prehensile interaction, quasi-static vs. dynamic interaction, terrain and payload conditions), and (iii) controller families (e.g., inverse-dynamics WBC, trajectory optimization/MPC, learning-based policies, and hybrid stacks). This triad is not only an organizing device; it functions as an *analytical lens* that enables cross-paradigm comparison under a common dynamic backbone and yields insights that are difficult to extract from prior surveys.

Concretely, the modeling–tasks–controls triad supports three insights that are often implicit or fragmented in existing reviews:

- **Modeling → feasibility boundaries:** identifying which task regimes become tractable (or ill-posed) under specific modeling choices, especially when contact transitions, frictional uncertainty, or object dynamics dominate performance.
- **Tasks → controller selection criteria:** translating task attributes (interaction type, dynamics, terrain, payload, sensing) into actionable ''when-to-use-what'' guidance, rather than a descriptive taxonomy of methods.
- **Controls → shared trade-offs and failure modes:** exposing cross-family trade-offs (computation vs. robustness, constraint satisfaction vs. adaptability, compliance vs. precision) and recurring failure modes (contact mis-modeling, delayed feedback, underactuated base coupling) across WBC, MPC, learning, and hybrid approaches.

Mobile manipulation is central to tasks that require both mobility and precise physical interaction, including tool delivery, inspection with intervention, and material handling in human-centered workspaces [3]. These systems must integrate perception, mobility, and dexterity under tight real-time constraints, especially when interaction forces and environmental uncertainties couple locomotion and manipulation. A representative example is Boston Dynamics' Spot robot equipped with a manipulator arm (Fig. 1), illustrating how legged platforms can execute manipulation tasks while remaining operationally mobile in industrial and field contexts.

Quadruped robots equipped with arms are increasingly relevant in manufacturing, defense, agriculture, and construction, where mobility over uneven terrain and physical interaction are both required. In manufacturing and inspection, such systems can reduce downtime by accessing constrained workspaces and performing intervention actions (e.g., handling parts, operating fixtures). In defense and hazardous operations, quadruped platforms can support reconnaissance and manipulation of risky objects while maintaining personnel standoff. In agriculture and outdoor infrastructure, physical interaction (e.g., sampling, grasping, cutting, or tool use) can substantially expand task coverage beyond locomotion-only inspection.

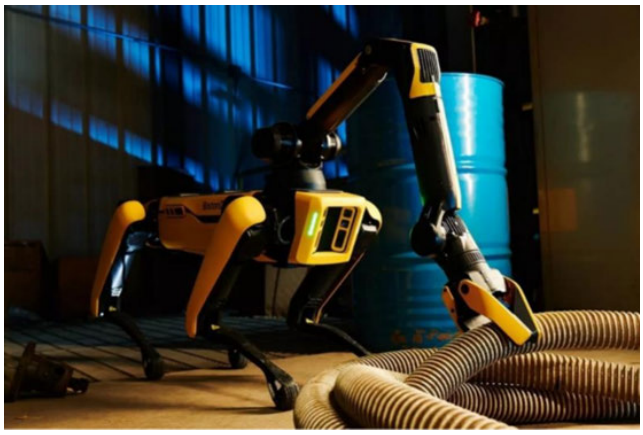


FIGURE 1. Boston Dynamics' Spot performing a loco–manipulation task in a dynamic environment [4].

Quadrupeds have attracted sustained attention due to agility and terrain adaptability relative to wheeled or tracked platforms, particularly in unstructured environments. However, stable loco–manipulation raises fundamental difficulties during dynamic configuration changes and contact transitions. Key challenges include:

- **Stability under interaction:** maintaining dynamic stability while manipulation forces and base motions shift the center of mass and ground reaction forces; stability constraints depend on stance, contact configuration, and interaction direction.
- **Reliable and safe contact:** achieving secure prehensile or non-prehensile interaction without slip, loss of grasp, or object damage, while locomotion-induced vibrations and terrain compliance perturb contact conditions.
- **Bidirectional coupling effects:** locomotion and manipulation strongly influence each other through whole-body dynamics, contact constraints, and payload effects; compliant legs can improve disturbance rejection but degrade end-effector precision, motivating contact-aware whole-body coordination and adaptive feedback.

Recent literature exemplifies the breadth of approaches. Optimization-based whole-body control has been used to coordinate arm and base through inverse-dynamics WBC formulations for simultaneous stability and manipulation [3]. Learning-based approaches have demonstrated unified whole-body policies that capture arm–leg coupling and address sim-to-real gaps via online adaptation [2]. Robust trajectory optimization has incorporated object dynamics

into whole-body models to improve disturbance tolerance across contact changes [5]. Continual learning has also been explored to mitigate catastrophic forgetting in sequential skill acquisition for legged platforms [6].

Broadly, three methodological trends can be identified. First, *model-based and optimization-based* approaches (e.g., MPC/WBC) exploit explicit kinematic and dynamic structure and are effective for contact-rich regimes, but can require significant modeling effort and may degrade under unmodeled variability [1], [5], [7]. Second, *learning-for-command* pipelines use learning to generate high-level commands that are executed by inverse kinematics or built-in controllers, which can simplify deployment but often underutilize whole-body coordination and may yield impractical Inverse Kinematics (IK) in constrained scenarios [8], [9]. Third, *end-to-end deep RL* has enabled whole-body control policies; early systems combined learned locomotion with model-based arms [10], while later work explored unified policies for end-effector tracking and more expressive task-space objectives [11], [12]. The modeling–tasks–controls triad adopted in this survey makes these categories *commensurable* by explicitly tying each family to its modeling assumptions and the task regimes where it is most effective.

Beyond technical progress, quadruped loco–manipulation aligns with *Industry, Innovation, and Infrastructure* (SDG 9) by enabling safer inspection, maintenance, and intervention in hard-to-reach assets such as pipelines, bridges, and refineries [13], [14]. Advances in floating-base modeling, contact-aware planning, and whole-body control support contact-rich procedures (e.g., valve turning, door operation, debris clearing) on uneven terrain, reducing human exposure to hazards and improving asset availability [13]. In parallel, *Sustainable Cities and Communities* (SDG 11) is supported by systems capable of traversing rubble, operating fixtures in damaged buildings, and delivering tools or supplies; teleoperation and shared autonomy can improve safety and situational awareness in disaster response settings [15], [16].

In this context, the survey organizes the field around a modeling–tasks–controls triad, provides "when-to-use-what" decision matrices for selecting control families under terrain/payload/sensing constraints, and emphasizes benchmarking and reproducibility through simulation and digital-twin pipelines.

The remainder of the review is organized as follows. Section II contrasts prior surveys and clarifies positioning within the broader mobile manipulation and legged locomotion literature. Section III introduces the mathematical modeling framework for quadruped robots with manipulators, used as a common backbone across controller families. Section IV proposes a task taxonomy specific to loco–manipulation (interaction type, dynamics, terrain, payload, coordination, and sensing). Section V reviews control methods, organized into model-based, optimization-based, learning-based, and hybrid families, with impedance/force control as a cross-cutting layer. Section VI discusses whole-body coordination challenges that arise regardless of the chosen controller family. Section VII surveys simulation environments and digital-twin platforms that support development and benchmarking. Section VIII summarizes trends and trade-offs, and Section IX concludes with open research directions for quadruped loco–manipulation. Figure 2 provides a tree-style overview connecting modeling, task taxonomy, control methods, coordination challenges, and simulation/digital-twin aspects.

II. RELATED SURVEYS AND OUR POSITIONING

To clarify how this survey differs from prior reviews, Table 1 contrasts scope and depth across key dimensions: (i) floating–base/contact modeling for legged systems, (ii) a task taxonomy *specific* to loco–manipulation (interaction type, terrain, payload, coordination, sensing), (iii) a control taxonomy spanning WBC/MPC/RL/hybrids, and (iv) treatment of benchmarks & reproducibility. However, our key distinction is that we do not treat these dimensions independently. Instead, we use the modeling–tasks–controls triad as an explicit *linking structure* that turns coverage into *actionable insight*: modeling assumptions delimit feasible task regimes, task attributes induce controller requirements, and controller families expose consistent trade-offs and failure modes across the literature.

A. WHAT PRIOR REVIEWS COVER

Broad mobile–manipulation overviews (e.g., [17], [18]) summarize platforms and challenges with helpful taxonomies, but provide only partial coverage for floating–base contact modeling and offer limited guidance on *method selection* for legged contact tasks. Quadruped–manipulator synopses (e.g., [19]) compile hardware and representative systems but remain high-level on task decomposition and control classification. WBC tutorials [20] are authoritative on impedance/passivity and stack-of-tasks, yet do not center the loco–manipulation problem (co-optimizing leg contacts and arm interaction) or provide decision criteria across MPC/RL/hybrid stacks. Legged–locomotion surveys (e.g., [21]) treat modeling and control deeply, but manipulation *while moving* is largely out of scope.

B. NEWLY RELEVANT SURVEYS

Two recent reviews further motivate our positioning: (i) Tang et al. survey deep RL for *robotics* with an emphasis on *real-world deployments* [22], offering breadth on learning but little on legged contact modeling or quadruped–arm integration; and (ii) a 2024 Heliyon review synthesizes control techniques for *legged locomotion* [23], again comprehensive for locomotion but without a quadruped–arm task taxonomy or benchmark guidance. These additions reinforce the gap our paper fills.

C. WHAT NEW INSIGHTS EMERGE FROM THE TRIAD

By coupling modeling, tasks, and controls explicitly, the triad enables insights that are difficult to extract from prior survey structures. First, it reveals *mismatches* between

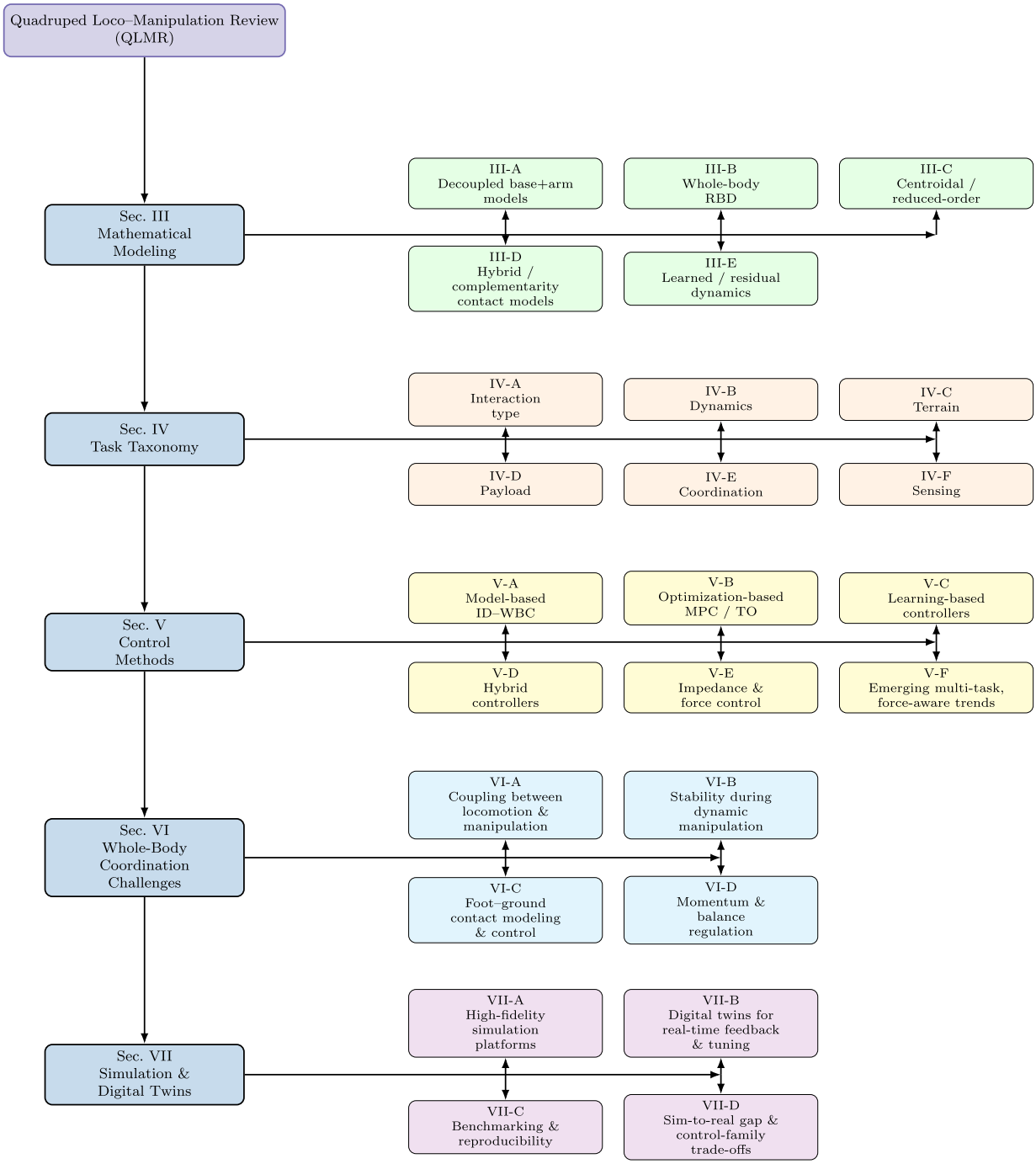


FIGURE 2. Tree-style roadmap of the survey. The main sections (left column) form a vertical chain, and each section branches to its subsections (right) describing modeling, tasks, control, whole-body loco–manipulation, and simulation/digital-twin aspects.

modeling fidelity and task demands (e.g., when neglecting object dynamics or contact compliance undermines forceful interaction or rapid contact switching). Second, it supports *actionable selection guidance* by mapping task attributes (interaction type, dynamics, terrain, payload, sensing) to controller requirements (constraint handling, contact consistency, bandwidth, computational budget), yielding ''when-to-use-what'' decision criteria rather than descriptive method catalogs. Third, it enables *cross-family synthesis* by making trade-offs and failure modes comparable across WBC, MPC, learning-based, and hybrid stacks (e.g., computation vs. robustness, compliance vs. precision, constraint satisfaction vs. adaptability).

D. OUR CONTRIBUTION AND POSITIONING

This review centers quadruped loco–manipulation and contributes a balanced modeling–tasks–controls triad with: (a) a reproducible task taxonomy mapped to representative

TABLE 1. Prior reviews vs. this work (✓ Covered, ● partially Covered ✗ Not Covered). We state each paper's *scope* as declared by the authors, then indicate coverage depth per column.

Review (Year)	Scope (as stated)	Quad+Arm	Modeling	Task taxonomy	Control taxonomy	Benchmarks & repro-ducibility	What's missing (we add)
[17] (2020)	Broad *mobile manipulation*: platforms, challenges, app domains	✗	●	●	●	✗	Quadruped-specific contact modeling; legged method selection; reproducibility guidance
[19] (2023)	*Quadruped manipulators* overview: hardware & use-cases	✓	●	●	●	✗	Balanced modeling-task-control triad; decision matrix; benchmarks
[20] (2016)	*Whole-body impedance/control* (tutorial/survey)	●	✓	✗	●	✗	Extend WBC to loco-manip with contact feasibility (WBC/MPC hybrids) and task axes
[18] (2022)	*Mobile manipulation* methods (platform-agnostic)	✗	●	●	●	✗	Floating-base contact modeling for quadrupeds; legged method selection guidance
[21] (2022)	*Legged locomotion* control (optimization-centric)	✗	✓ (locomotion)	● (locomotion)	● (locomotion)	✗	Add manipulation-while-moving; cross-family comparisons incl. RL/hybrids
[22] (2024)	*Deep RL for robotics* with *real-world successes*	✗	✗	●	✓ (RL)	●	Quadruped contact modeling; Quad+Arm focus; integration with MPC/WBC and task axes
[23] (2024)	*Legged locomotion*: control techniques & applications	✗	✓ (locomotion)	● (locomotion)	●	✗	Loco-manipulation taxonomy; Quad+Arm modeling; reproducible benchmarks
This survey (2025)	**Quadruped *loco-manipulation*: modeling-tasks-controls**	✓	✓	✓	✓	✓	**Unified triad + critical comparison + "when-to-use-what" matrix + benchmarks & reproducibility**

scenarios; (b) a control classification integrating model-based (WBC/MPC), learning-based, and hybrid stacks, together with explicit "when-to-use-what" guidance under terrain, payload, and sensing constraints; and (c) benchmarking and reproducibility practices (simulation/digital-twin pipelines, datasets where available, ablation recommendations, and code availability reporting). Accordingly, this paper complements prior work by providing *legged contact modeling depth*, *task-axis standardization*, and *actionable method-selection guidance* for quadruped loco–manipulation.

Having positioned this review relative to prior surveys, we now turn to the floating-base and contact modeling assumptions that underlie most loco–manipulation controllers. Section III formalizes the dynamics and modeling abstractions that will be referenced throughout the remainder of the paper.

III. MATHEMATICAL MODELING

A critical question for loco-manipulation is not only *how* to model quadruped robots with arms, but also *which* modeling abstractions are actually used by the different control families surveyed later (Section V) and what this implies for their performance and limitations. In this section we therefore keep the mathematical formulation deliberately concise and explicitly annotate how each modeling choice underpins particular controller types (ID-WBC, MPC, RL, hybrids) and the coordination challenges discussed in Section VI.

In line with the conventions of the Pinocchio library [24], [25], we define the generalized coordinate vector $\mathbf{q}$ and the generalized velocity vector $\mathbf{v}$ for a quadruped robot equipped with a manipulator. The robot's state is expressed as:

$$\mathbf{q} = \begin{bmatrix} {}^{W}\mathbf{x}_b \\ \mathbf{q}_b \\ \mathbf{q}_j \end{bmatrix} \in SE(3) \times \mathbb{R}^n, \quad \mathbf{v} = \begin{bmatrix} {}^{W}\dot{\mathbf{x}}_b \\ {}^{B}\boldsymbol{\omega}_b \\ \dot{\mathbf{q}}_j \end{bmatrix} \in \mathbb{R}^{6+n}. \quad (1)$$

Here ${}^{W}\mathbf{x}_b \in \mathbb{R}^3$ is the position of the base in the world frame, $\mathbf{q}_b$ is the unit quaternion for base orientation, ${}^{W}\dot{\mathbf{x}}_b$ is the linear velocity of the base, and ${}^{B}\boldsymbol{\omega}_b$ is the angular velocity of the base expressed in the base frame. The joint configuration and velocities are represented by $\mathbf{q}_j, \dot{\mathbf{q}}_j \in \mathbb{R}^n$, where $n = 18$ includes 6 DoFs from the arm. The full floating-base dynamics (rigid body with contacts) are

$$\mathbf{M}(\mathbf{q})\dot{\mathbf{v}} + \mathbf{h}(\mathbf{q}, \mathbf{v}) = \mathbf{S}\boldsymbol{\tau} + \mathbf{J}_c^{\top}(\mathbf{q})\boldsymbol{\lambda} \quad (2)$$

subject to the contact constraint:

$$\mathbf{J}_c(\mathbf{q})\ddot{\mathbf{q}} + \dot{\mathbf{J}}_c(\mathbf{q})\dot{\mathbf{q}} = \mathbf{0} \quad (3)$$

In the above equations, $\mathbf{M}(\mathbf{q}) \in \mathbb{R}^{(6+n)\times(6+n)}$ is the inertia matrix, and $\mathbf{h}(\mathbf{q}, \mathbf{v}) \in \mathbb{R}^{6+n}$ includes Coriolis, centrifugal, and gravitational terms. The torque vector $\boldsymbol{\tau} \in \mathbb{R}^n$ applies only to actuated joints via the selection matrix $\mathbf{S} =$

$\left[\mathbf{0}_{6\times n}^\top \ \mathbf{I}_{n\times n}^\top\right]^\top$. $\mathbf{J}_c(\mathbf{q}) \in \mathbb{R}^{3m\times(6+n)}$ is the contact Jacobian, and $\boldsymbol{\lambda} \in \mathbb{R}^{3m}$ represents the contact forces. For analysis and control, (2) is often split into base and joint components,

$$\mathbf{M}_b\dot{\mathbf{v}} + \mathbf{h}_b = \mathbf{J}_{c,b}^\top\boldsymbol{\lambda}, \tag{4}$$

$$\mathbf{M}_j\dot{\mathbf{v}} + \mathbf{h}_j = \boldsymbol{\tau} + \mathbf{J}_{c,j}^\top\boldsymbol{\lambda}. \tag{5}$$

Here $\mathbf{M}_b, \mathbf{h}_b, \mathbf{J}_{c,b}$ correspond to the floating base, and $\mathbf{M}_j, \mathbf{h}_j, \mathbf{J}_{c,j}$ correspond to the actuated joints.

Equations (2)-(5) are more than textbook notation: they are exactly the constraints embedded in inverse-dynamics whole-body control (ID-WBC, Section V-A) and whole-body MPC (Section V-B). Works that solve a single optimization problem over joint accelerations, torques, and contact forces (e.g., [26], [27], [28]) simply instantiate different cost functions and horizons on top of this same backbone. As we will see, when controllers deviate from this full model (e.g., to a centroidal or decoupled abstraction), they gain computational efficiency but lose some expressiveness for coordinating locomotion and manipulation.

To enhance control, particularly for locomotion and manipulation, centroidal momentum dynamics can be used to express the system's motion relative to the center of mass (CoM). The rate of change of centroidal momentum is written as:

$$\dot{\mathbf{h}}_G = \mathbf{A}_G(\mathbf{q})\dot{\mathbf{v}} + \dot{\mathbf{A}}_G(\mathbf{q}, \mathbf{v})\mathbf{v} \tag{6}$$

Here $\dot{\mathbf{h}}_G = \left[\dot{\mathbf{l}}_G^\top \ \dot{\mathbf{k}}_G^\top\right]^\top \in \mathbb{R}^6$ includes both linear and angular momentum rates. $\mathbf{A}_G(\mathbf{q}) \in \mathbb{R}^{6\times(6+n)}$ is the centroidal momentum matrix, and $\dot{\mathbf{A}}_G(\mathbf{q}, \mathbf{v})$ captures its time variation. These matrices provide a mapping between generalized coordinates and the CoM dynamics. Given that:

$$\dot{\mathbf{l}}_G = m\ddot{\mathbf{p}}_{\text{CoM}} \tag{7}$$

We can derive joint commands that ensure CoM trajectory tracking. Lastly, we express the joint torques in terms of the dynamic quantities using the joint subsystem equation:

$$\boldsymbol{\tau} = \mathbf{M}_j\dot{\mathbf{v}} + \mathbf{h}_j - \mathbf{J}_{c,j}^\top\boldsymbol{\lambda} \tag{8}$$

Centroidal quantities such as $\mathbf{h}_G$ and $\ddot{\mathbf{p}}_{\text{CoM}}$ play a central role in optimization-based control (Section V-B) because they offer a reduced-order view in which whole-body feasibility is approximated via CoM and momentum balance. In practice, MPC formulations that operate on centroidal dynamics (e.g., [28], [29]) trade off some joint-level accuracy and arm/base coupling detail for longer horizons and faster solve times, which directly impacts their ability to handle the coordination challenges in Section VI (especially momentum and balance regulation).

Equations (2)-(8) are the standard constraints embedded in optimization-based controllers (e.g., Whole-Body Control (WBC), Model Predictive Control MPC), with (2) acting as the principal dynamics constraint and (3) enforcing contact consistency. This torque expression enables the use of optimization-based controllers, such as WBC or MPC, where Equation (4) often serves as the principal constraint. The formulation also supports embedding centroidal or CoM-based objectives within the control problem for dynamic, stable, and contact-aware behavior generation.



FIGURE 3. Schematic URDF model of the Unitree Go2 quadruped with a back-mounted manipulator. The base and arm follow the public and Go2+arm URDF integrations, aligned with vendor documentation for Go2 and the D1-T on-board robotic arm [30], [32], [33].

Figures 3-4 ground the platform and notation used in this review. Figure 3 shows a Unitree Go2 with a back-mounted manipulator as modeled in URDF, consistent with vendor documentation and public robot descriptions [30], [31], [32], [33]. Figure 4 summarizes the floating-base frames, leg contact points, and stance geometry that underpin our WBC/MPC formulations and Jacobian/centroidal notation throughout the paper [34], [35].

With this baseline in place, we now outline the main modeling *abstractions* that appear in loco-manipulation studies, each of which maps naturally onto different regions of the control taxonomy in Section V. Rather than treating them as purely theoretical variants, we emphasize which controllers they enable and what limitations they introduce.

Complementing the baseline floating–base formulation in (2)–(8), loco–manipulation studies typically adopt one of five modeling abstractions: (i) *decoupled base+manipulator models*, (ii) *whole–body rigid–body dynamics (RBD)* with a single floating base, (iii) *reduced–order/centroidal* models, (iv) *hybrid/complementarity* models that make contact modes explicit, and (v) *learned dynamics* (stand–alone or as residuals on top of RBD). Below we formalize each and connect them to control use-cases.

A. DECOUPLED BASE + MANIPULATOR MODEL

The legged base and the arm are modeled separately, with limited or scheduled coupling. A common choice is a

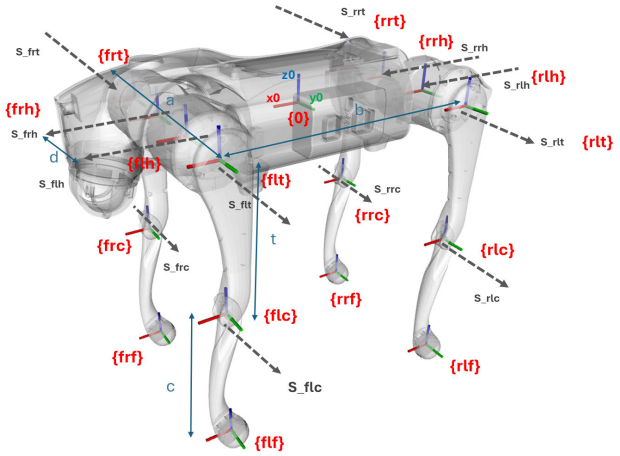


FIGURE 4. Quadruped modeling schematic showing the floating-base frame {0}, body axes (x_0, y_0, z_0), leg frames, and typical contact points used to define Jacobians, centroidal momentum, and friction-cone constraints in the loco–manipulation controllers reviewed here [34], [35].

reduced model for the base (e.g., centroidal or linearized base dynamics) and a rigid–arm model in task or joint space. Writing feet contacts as external wrenches on the base and the end–effector wrench on the arm,

$$M_b(\xi_b)\dot{v}_b + h_b(\xi_b, v_b) = \sum_{i=1}^{m} {}^{B}X_{F_i}^{\top} f_{F_i} + {}^{B}X_{E}^{\top} w_E, \tag{9}$$

$$M_a(q_a)\ddot{q}_a + C_a(q_a, \dot{q}_a)\dot{q}_a + g_a(q_a) = \tau_a + J_E^{\top}(q) f_E, \tag{10}$$

where ξ_b is a minimal base pose, v_b the base spatial velocity, f_{F_i} the i-th foot wrench, and $w_E = -{}^{B}X_{E}^{\top} f_E$ is the mount wrench equal and opposite to the end–effector wrench f_E.[1] The arm can be commanded by inverse kinematics/impedance while the base controller regulates momentum/CoM and allocates ground reaction forces. This lightweight factorization is used in practice for learned-locomotion + model-based manipulation pipelines [10] or for task-level planning where tight dynamic coupling is secondary.

From a control perspective, decoupled models bias designs toward *modular* stacks: a learned or model-based locomotion controller (often centroidal or reduced-order) is paired with a separate arm controller (IK or impedance). This simplifies software and tuning, but makes it harder to address strong base-arm coupling and dynamic manipulation challenges (Section VI), especially when payloads are heavy or the arm induces large base disturbances.

B. WHOLE–BODY FLOATING–BASE RBD (SINGLE MODEL)

Your equations (2)–(8) define the standard whole–body RBD with contacts, directly suitable for inverse–dynamics WBC and WB–MPC [26], [27]. For completeness, the partition used by many WBC stacks writes

$$\begin{bmatrix} M_{bb} & M_{bj} \\ M_{jb} & M_{jj} \end{bmatrix} \begin{bmatrix} \dot{v}_b \\ \ddot{q}_j \end{bmatrix} + \begin{bmatrix} h_b \\ h_j \end{bmatrix} = \begin{bmatrix} 0 \\ \tau \end{bmatrix} + \begin{bmatrix} J_{c,b}^{\top} \\ J_{c,j}^{\top} \end{bmatrix} \lambda, \tag{11}$$

$$J_c \begin{bmatrix} v_b \\ \dot{q}_j \end{bmatrix} = 0 \text{ (no–slip)}, \tag{12}$$

with torque and acceleration limits and friction cones added as in (3)–(8).

This is the modeling choice behind most ID-WBC controllers and many WB-MPC implementations. Its main advantage is that it directly supports multi-priority task hierarchies and explicit enforcement of joint, torque, and friction constraints (critical for the coordination challenges in Table 22), but it also imposes the heaviest computational and modeling burden. Consequently, works targeting highly dynamic behaviors on embedded hardware either approximate this model (e.g., centroidal) or restrict horizon length in MPC, which partially explains the performance gaps we highlight in Section V.

C. REDUCED–ORDER/CENTROIDAL MODELS

A centroidal abstraction keeps only the center–of–mass (CoM) and total momentum, discarding joint–level details [35]. The Newton–Euler balance is

$$\underbrace{\dot{h}_G}_{[\dot{\ell}_G;\, \dot{k}_G]} = \sum_{i=1}^{m} \begin{bmatrix} I \\ (p_{F_i} - p_G)^{\wedge} \end{bmatrix} f_{F_i} + \begin{bmatrix} mg \\ 0 \end{bmatrix} + w_{\text{ext}}, \tag{13}$$

$$\dot{\ell}_G = m\ddot{p}_G, \tag{14}$$

where p_G is the CoM position, ℓ_G and k_G are the linear and angular momentum about the CoM, $(\cdot)^{\wedge}$ the skew operator, and w_{ext} external wrenches. The centroidal momentum matrix $A_G(q)$ used in (6) gives $h_G = A_G(q)v$ and is useful to couple reduced control costs to whole–body kinematics (e.g., in WB–MPC).

Table 2 summarizes how modeling abstractions tend to align with controller families; here we add a concrete example showing how that choice manifests as measurable experimental trade-offs. In [1], the authors report a dynamic lifting loco-manipulation experiment and explicitly evaluate reduced-order MPC models. They show that a centroidal-dynamics model augmented with object dynamics enables substantially more aggressive reference execution and faster settling than simpler template alternatives (e.g., a single-rigid-body template), while maintaining real-time solvability. in their reported study, the centroidal+object formulation supports a minimum reference lifting time of 0.2 s with a settling time of 0.77 s, whereas removing object dynamics or using a simpler template leads to markedly slower transients (reported settling times on the order of multiple seconds), with comparable average MPC computation times (tens of milliseconds). This illustrates the core trade-off behind selecting centroidal dynamics over more detailed whole-body models in online MPC: centroidal dynamics captures system momentum exchange and

[1] ${}^{B}X_{(\cdot)}$ denotes the spatial transform from the respective contact/EE frame to the base frame.

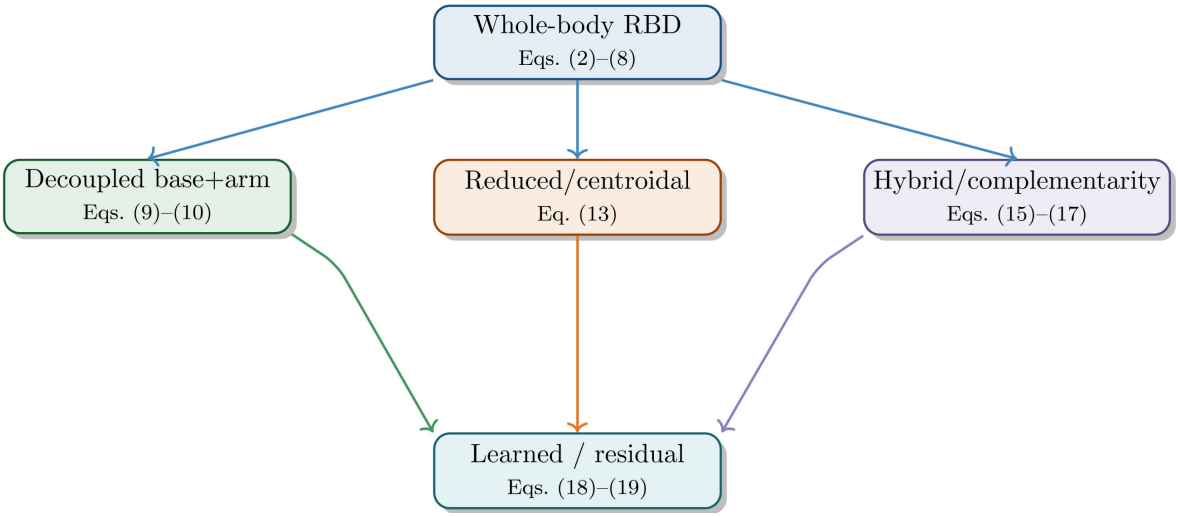


FIGURE 5. Modeling abstractions and their relations (color-coded). Controllers select the simplest model that preserves safety/feasibility: whole-body RBD for WBC/WB-MPC; centroidal for CoM/momentum control; complementarity when mode changes (stick/slip/impact) must be optimized; and learned/residual models to capture unmodeled effects.

TABLE 2. Modeling choices and typical usage in loco–manipulation.

Model	State/contacts	Typical controller	Pros / Limits (examples)
Decoupled base+arm	Base reduced; arm joint model; contacts as wrenches	IK/impedance for arm; centroidal/MPC for base	Simple and modular; limited tight coupling [10]
Whole-body RBD	$[q, v]$ full; J_c, λ	ID-WBC; WB-MPC	Exact constraints; higher compute [26], [27]
Centroidal	(p_G, h_G); GRFs	CoM/momentum MPC	Efficient, contact-feasible; loses joint detail [35]
Hybrid/complementarity	Modes explicit; impulses	Contact-implicit TO/MPC	Finds contact schedules; nonsmooth/expensive [36]
Learned / residual	x or features; learned contact terms	Model-Based RL (MBRL); residual MPC/WBC	Captures unknowns; data needs [37]–[39]

contact-force redistribution (improving transient whole-body behavior during manipulation), but it does not directly enforce joint-level actuation feasibility and full multibody dynamics. Accordingly, [1] uses a torque-level tracking layer (inverse-dynamics whole-body control / impedance shaping) beneath the MPC to realize the plan and handle joint/actuator limits. At the same time, the same study notes that moving the optimization model toward higher-fidelity whole-body formulations (e.g., full centroidal plus full kinematics, closer to full multibody RBD in the OCP) can prevent real-time solution at the desired rates, motivating the centroidal abstraction for the online layer.

Reduced-order models are the workhorse of many optimization-based controllers in Section V-B, particularly when dynamic locomotion and coarse manipulation must be handled over long horizons. They excel at addressing some coordination challenges (e.g., CoM/momentum regulation under disturbances), but their limited representation of joint and contact geometry can make fine arm tasks and precise force distribution harder to encode, which often motivates a hybrid reduced+full model stack.

Many pipelines plan contact forces and CoM/momentum evolution using a centroidal abstraction, then realize the plan through a full-body tracking layer (e.g., ID-WBC). Let the centroidal planner use an approximate mapping $W(q)$ from stacked contact forces f to centroidal momentum rate,

$$\dot{h}_G \approx W(q)f + w_{\text{ext}}, \tag{15}$$

whe the true relationship is

$$\dot{h}_G = W(q)f + w_{\text{ext}}. \tag{16}$$

The resulting mismatch enters as an effective disturbance on the floating base:

$$\Delta\dot{h}_G := \dot{h}_G - \dot{h}_G^{\star} = \big(W - \tilde{W}\big)f^{\star} + \big(w_{\text{ext}} - \tilde{w}_{\text{ext}}\big), \tag{17}$$

which the tracking controller must compensate using feasible contact forces and joint torques. When compensation is limited by friction cones, unilateral constraints, or actuator

saturation, the mismatch manifests as base-state tracking error. Since the end-effector (EE) pose depends on both base and joint configuration, the induced EE error follows (first order)

$$\delta x_{\mathrm{EE}} \approx J_{\mathrm{EE},b}\,\delta q_b \;+\; J_{\mathrm{EE},j}\,\delta q_j, \tag{18}$$

so base drift/attitude error caused by centroidal mismatch directly degrades manipulation accuracy even if the arm joints track well. This clarifies why reduced-order planning can remain effective for dynamic balance yet require a strong full-body realization layer (ID-WBC/impedance) to recover task-space accuracy under payload changes or strong interaction.

D. HYBRID/COMPLEMENTARITY CONTACT MODELS

Hybrid models make contact modes explicit and use complementarity to encode normal contact and stick–slip [36], [40], [41]. For a foot with gap $\phi(q) \geq 0$ and normal force $\lambda_n \geq 0$,

$$\phi(q) \geq 0, \quad \lambda_n \geq 0, \quad \phi(q)\,\lambda_n = 0. \tag{19}$$

Tangential no–slip with Coulomb friction (μ) can be written via velocity complementarity:

$$\|\lambda_t\| \leq \mu\,\lambda_n, \qquad 0 = v_t \perp \lambda_t \in \partial\|\cdot\|, \tag{20}$$

Here, $\phi(\mathbf{q}) \geq 0$ denotes the non-penetration gap function at the contact point, $\lambda_n \geq 0$ is the normal contact force, $\boldsymbol{\lambda}_t \in \mathbb{R}^2$ is the tangential contact force, $\mathbf{v}_t$ is the tangential relative velocity at the contact, μ is the Coulomb friction coefficient, the symbol $\perp$ indicates standard complementarity (i.e., $a \perp b \Leftrightarrow a \geq 0,\ b \geq 0,\ a^\top b = 0$), and $\partial\|\cdot\|$ denotes the subgradient of the Euclidean norm,

and impacts are handled with an impulse–momentum update

$$M(q)\,(v^+ - v^-) = J_{\mathrm{imp}}^\top \Lambda, \qquad 0 \leq \Lambda_n \perp \phi(q) \geq 0, \tag{21}$$

with Λ the impact impulses. Such contact-implicit models appear in pushing/manipulation MPC and trajectory optimization when the schedule of contacts is not known *a priori* [36], [42], [43].

Contact-implicit models are most tightly connected to non-prehensile and highly dynamic tasks in our taxonomy (Section IV) and are primarily used within optimization-based planners (Section V-B) rather than real-time WBC loops. They can address coordination challenges related to contact scheduling (e.g., when and where to create or break foot or hand contacts), but their nonsmooth structure and high computational cost limit their use to offline or low-rate planning, which then has to be tracked by simpler controllers.

A core practical trade-off is whether contact modes are (i) fixed/scheduled and enforced via unilateral/friction constraints in a QP/MPC layer, or (ii) implicitly decided by the optimizer through complementarity/contact-implicit constraints such as $0 \leq \phi(q) \perp \lambda_n \geq 0$. Scheduled-contact formulations typically yield convex QPs (or structured SQPs) with strong real-time behavior and are widely used in locomotion MPC stacks [44], [45]. Contact-implicit complementarity models can better represent uncertain contact switching and stick–slip effects, but they introduce nonsmooth MPCC/NLP structure that can increase sensitivity to initialization and solver tuning [36]. Recent work demonstrates that carefully structured contact-implicit NMPC can nevertheless reach high update rates on quadrupeds (reported solver rates up to ∼190 Hz) [46], but such approaches remain comparatively complex and are still less common for high-DoF loco-manipulation (quadruped+arm) where contact, arm dynamics, and perception delays interact.

E. LEARNED DYNAMICS MODELS

Learned models either *replace* $f(x,u)$ or *augment* physics with a residual:

$$\dot{x} = f_\theta(x,u) \quad \text{(pure learned)}, \tag{22}$$

$$\dot{x} = f_{\mathrm{RBD}}(x,u) + \Delta_\theta(x,u) \quad \text{(residual on RBD)}, \tag{23}$$

where $x = [q; v]$. Parameters θ are trained from rollouts $\{x_k, u_k\}_{k=0}^{T}$ by minimizing a simulation or one–step prediction loss (e.g., negative log-likelihood for GPs in PILCO [37], supervised rollout loss for neural dynamics [38], or physics–structured losses as in Deep Lagrangian Networks [39]). Classical identification of inertial parameters via regressor $Y(q,\dot{q},\ddot{q})$ is also used to refine f_{RBD} [47]. In loco–manipulation, learned Δ_θ often captures unmodeled contacts, compliance, or payload effects while preserving constraint handling in WBC/MPC.

These learned models are the backbone of model-based RL (Section V-C) and hybrid MPC+learning approaches (Section V-D). They are particularly attractive for coordination challenges involving unknown payloads, complex contact friction, or wear-and-tear (Table 22), where hand-crafted models are brittle. However, they introduce their own failure modes (model bias, data dependence) that directly influence controller selection in Sections V and VI.

In summary, Table 2 can be read as a bridge between modeling and control: each row corresponds to a different slice of the control taxonomy (WBC on whole-body RBD, MPC on centroidal or hybrid models, learning on residual or pure learned dynamics). The rest of the paper repeatedly returns to this mapping when assessing how well each controller family addresses task axes (Section IV) and coordination challenges (Section VI).

The modeling choices summarized above answer the question of *how* quadruped loco-manipulation platforms are represented at the dynamics level. In the next section we shift to *what* these robots are asked to do by introducing a task taxonomy for loco-manipulation; this taxonomy will later be cross-referenced with the control families in Section V.

IV. TASK TAXONOMY FOR LOCO-MANIPULATION

To compare loco-manipulation controllers across studies, we introduce a six-axis task taxonomy that we use

consistently throughout the review. Rather than choosing these axes ad hoc, we derived them in two steps. First, we surveyed existing taxonomies for legged locomotion, mobile manipulation, and human-robot collaboration [17], [19], [21], collecting candidate attributes that repeatedly appear in experimental sections (e.g., terrain, payload, interaction type). Second, we manually coded the tasks reported in the loco-manipulation literature corpus using this candidate set and iteratively removed attributes that were either rarely reported, strongly redundant with others, or weakly correlated with controller design choices. We retained six axes-interaction type, dynamics, terrain, payload, coordination, and sensing-because (i) they can be inferred from nearly all experimental descriptions, (ii) each axis is associated with distinct control requirements (e.g., stability margins vs. perception robustness), and (iii) changing a single axis while holding the others fixed typically changes the dominant controller family used in practice (as summarized in Table 4). In this sense, the axes are ''approximately orthogonal'' from a control-design standpoint: they separate the main sources of difficulty without pretending to be mathematically independent.

To improve reproducibility, the six-axis taxonomy was derived via an open-coding workflow commonly used in systematic mapping and qualitative synthesis [48], [49], [50]. The process followed three stages. First, candidate task attributes were collected from existing taxonomies and surveys in legged locomotion, mobile manipulation, and HRI. Second, tasks described in the loco-manipulation corpus were manually coded using a shared codebook (axis definitions and inclusion rules), and the codebook was iteratively refined through pilot passes and adjudication. Third, attributes were pruned when they were rarely reported, strongly redundant with others, or weakly connected to controller design decisions in the corpus. Algorithm 1 summarizes the procedure; the final codebook and the coded paper–axis matrix are provided as supplementary material.

To further reinforce the empirical grounding of the taxonomy, we report how the coded corpus populates each axis level. Table 3 summarizes the distribution of the coded corpus across the six axes (paper-level counts and percentages). For reproducibility, we also provide the underlying coded paper-by-axis spreadsheet (machine-readable supplementary material), including evidence links used during coding. In this revision, we report statistics for $N = 15$ loco–manipulation papers that explicitly demonstrate legged loco–manipulation tasks (locomotion and manipulation jointly), excluding locomotion-only method references that do not instantiate manipulation tasks.

The axes are intended to separate *distinct sources of control difficulty* rather than to be mathematically independent. To minimize overlap, each axis is defined by a single operational question: (i) **Interaction type:** which physical contact modality dominates (prehensile vs. non-prehensile)? (ii) **Dynamics:** are inertial effects dominant (quasi-static vs. dynamic)? (iii) **Terrain:** what is the expected variability /uncertainty of footholds and friction? (iv) **Payload:** what disturbance magnitude is injected by carried/manipulated loads? (v) **Coordination:** how many decision-makers share authority (single robot vs. human/teleop/multi-robot)? (vi) **Sensing:** what observation channels are available for feedback (proprio-only vs. exteroception such as vision)? For clarity, *interaction type* encodes the physical contact regime, whereas *sensing* encodes observability and feedback bandwidth; prehensile and non-prehensile tasks may be executed with either proprioception alone or with vision/perception in the loop. With these operational definitions, the axes are approximately orthogonal in a practical, control-design sense: each captures a different dominant bottleneck (constraints, uncertainty, compute, or observability) that can change controller feasibility or preferred architectures.

The retained axes were validated using three checks. First, **coverage**: each axis can be inferred from the vast majority of experimental task descriptions in the corpus, whereas pruned attributes were inconsistently reported. Second, **redundancy screening**: attributes that were strongly co-reported were merged or removed to avoid duplicating the same factor under different names (a standard step in coding-based synthesis [49]). Third, **controller relevance**: the final axes are those most consistently associated with differences in controller design choices across the corpus (e.g., shifts between WBC/impedance, WB-MPC/MPC variants, learning-based methods, and hybrid stacks), which motivates using the taxonomy to interpret the method trade-offs summarized in Table 4.

The taxonomy is thus not only descriptive but *control-driven*: each axis captures a factor that measurably influences which controllers are viable or advantageous. An overview with representative instances and controller families appears

Algorithm 1 Bottom-up Coding Procedure Used to Derive the Six-Axis Task Taxonomy

Require: Literature corpus $\mathcal{P}$; seed attributes $\mathcal{A}_0$; initial codebook $\mathcal{C}_0$

Ensure: Final axes $\mathcal{A}$; final codebook $\mathcal{C}$

1: **Seed extraction:** assemble candidate attributes $\mathcal{A}_0$ from prior taxonomies/surveys; define operational codes $\mathcal{C}_0$
2: **Pilot coding:** independently code a subset $\mathcal{P}_{\text{pilot}} \subset \mathcal{P}$ using $\mathcal{C}_0$
3: **Adjudication:** resolve disagreements and refine decision rules ($\mathcal{C}_0 \rightarrow \mathcal{C}_1$)
4: **Full coding:** code all tasks in $\mathcal{P}$ using $\mathcal{C}_1$, yielding a paper–attribute matrix $\mathbf{Z}$
5: **Coverage pruning:** remove attributes with low reporting coverage in experimental descriptions
6: **Redundancy pruning:** merge/remove attributes with strong redundancy (high co-occurrence/association)
7: **Control-affinity screening:** retain attributes that correspond to observable shifts in controller family usage
8: **Finalize:** output $\mathcal{A}$ and $\mathcal{C}$

TABLE 3. Distribution of the coded loco–manipulation corpus across the six taxonomy axes (N=15 papers). Percentages are reported in parentheses. "NR" indicates that the corresponding axis level was not explicitly reported in the source and was therefore left unspecified in the coded spreadsheet.

Axis	Level	# in bin
Interaction	Prehensile	8 (53.3%)
	Non-prehensile	7 (46.7%)
	NR	0 (0.0%)
Dynamics	Static / Quasi-static	0 (0.0%)
	Dynamic	15 (100.0%)
	NR	0 (0.0%)
Terrain	Flat	0 (0.0%)
	Uneven	5 (33.3%)
	Compliant / uncertain	0 (0.0%)
	NR	10 (66.7%)
Payload	Light ($< 5\%$)	0 (0.0%)
	Medium (5–15%)	2 (13.3%)
	Heavy ($> 15\%$)	1 (6.7%)
	NR	12 (80.0%)
Coordination	Single robot	10 (66.7%)
	Human–robot	1 (6.7%)
	Teleoperation	2 (13.3%)
	Multi-robot (rare)	2 (13.3%)
	NR	0 (0.0%)
Sensing	Proprio-only	11 (73.3%)
	Proprio + vision	4 (26.7%)
	NR	0 (0.0%)

in Table 4, and the remainder of this subsection explains why each axis is salient.

A. INTERACTION TYPE

Prehensile tasks involve grasping/holding (e.g., pick-and-place on uneven terrain [51], human–robot handover [52]). *Non-prehensile* tasks involve exchanging momentum without a grasp (e.g., door/handle pushing and collision-aware whole-body motion during contact-rich maneuvers [53]).

This axis distinguishes tasks where force closure and grasp stability dominate (prehensile) from those where transient contact forces and impact handling dominate (non-prehensile), which in turn favors different controller families (impedance/WBC vs. predictive MPC with explicit contact modeling).

B. DYNAMICS

Static/quasi-static interactions emphasize force regulation and posture with minimal base accelerations, typically handled by impedance/WBC controllers [3]. *Dynamic* interactions include momentum exchange and coordinated base–arm motion while moving; representative approaches include holistic/whole-body MPC for dynamic locomotion and manipulation [1] and unified-policy RL for whole-body coordination [2].

This axis captures how strongly inertial and coupling effects matter: static tasks can often be stabilized with myopic controllers, whereas dynamic tasks require trajectory optimization, predictive control, or policies trained on rich motion data.

C. TERRAIN

Flat terrain simplifies contact sequencing. *Uneven* terrain (steps, rubble) stresses footprint selection and robustness (e.g., ANYmal-B pick-and-place on rough terrain [51]). *Compliant* or uncertain terrain motivates compliance/impedance and contact adaptation [3].

Terrain thus acts as a proxy for contact uncertainty and model mismatch: moving from flat to uneven or compliant terrain shifts the bottleneck from manipulation kinematics to contact planning, robustness, and adaptation, and is one of the main drivers for choosing MPC or learning-based controllers over purely kinematic WBC.

D. PAYLOAD

We categorize payload as a fraction of robot mass: *light* $< 5\%$, *medium* 5-15%, *heavy* $> 15\%$. Heavy payload manipulation requires explicit GRF distribution and stability margins (e.g., HyQ+HyArm [54]); medium payloads often benefit from coordination between learned locomotion and model-based manipulation [10].

Payload directly affects feasible contact forces and stability margins. In our corpus, increasing payload while holding other axes fixed systematically pushes designs toward controllers that optimize GRF distribution (MPC/WBC with inequality constraints) or incorporate explicit disturbance modeling.

E. COORDINATION

Single-robot loco–manipulation covers most works (e.g., holistic MPC [1]). *Human–robot collaboration* introduces ergonomics and manipulability optimization [52]. *Teleoperation* scenarios (mixed reality) emphasize situational awareness and shared autonomy [15]. *Cooperative multi-robot* remains rare in quadrupeds; we reference broader mobile manipulation surveys for multi-agent settings [17].

This axis isolates how many decision-makers are in the loop and how authority is shared (autonomy vs. teleoperation vs. cooperation), which has a direct impact on the need for impedance shaping, shared-control schemes, and distributed planning.

F. SENSING

Proprio-only pipelines rely on state estimation and force sensing (typical in MPC/WBC [1]). *Proprio+vision* integrates RGB-D or learned perceptual features; examples include RL with unified whole-body policies using visual inputs and sim-to-real strategies [2], and MR-based teleoperation with rich perception [15].

Sensing breadth is separated as an axis because it conditions which controllers are even applicable: vision-conditioned RL and perception-in-the-loop MPC presuppose

dense exteroception, whereas many classical WBC stacks are designed for proprio-only operation.

Overall, Table 4 shows that the six axes expose clear controller affinities. Prehensile interaction on rough terrain at medium or heavy payloads tends to favor whole-body control with task-space impedance (often inside a QP) or whole-body MPC, with RL contributing grasp or sequencing primitives [1], [2], [51], [52]. Non-prehensile pushing/dragging, especially in dynamic regimes, is more commonly addressed by MPC/WBC formulations that enforce contact feasibility and collision avoidance [1], [53]. Uneven or compliant terrain shifts preference toward contact-planning MPC plus robust WBC or adaptation layers [3], [51], while increasing payload shifts solutions toward GRF-optimized or stability-margin controllers [54]. Coordination and sensing axes further differentiate settings where hierarchical optimization with compliant WBC is required for human-robot handover [52] from those where shared-control teleoperation or vision-conditioned RL is more appropriate [2], [15]. By grounding the taxonomy in these observed control-design transitions, we provide a more rigorous and actionable foundation than a purely descriptive categorization.

Having defined this six-axis task space and its controller affinities, we now revisit the main families of loco-manipulation controllers. Section V analyzes model-based, optimization-based, learning-based, and hybrid methods in terms of how well they address different regions of this taxonomy.

V. CONTROL METHODS

Designing reliable controllers for quadruped *loco-manipulation* is challenging: the system is high-dimensional and underactuated (floating base), contacts are intermittent and hybrid, and arm motions inject disturbances that must be balanced by the legs. Prior work tackles these issues with four main families of methods: (i) *model-based* controllers (inverse-dynamics whole-body control and related stacks), (ii) *optimization-based* methods (primarily MPC), (iii) *learning-based* policies, and (iv) *hybrid* combinations. Impedance/force-control layers and multi-modal perception (vision, force/torque, proprioception) cut across these families and are often embedded inside the main control loop rather than forming stand-alone classes [17], [19], [21].

This section is organized into five parts. Section V-A covers model-based inverse-dynamics whole-body control (WBC) and related stacks. Section V-B discusses optimization-based methods with a focus on model predictive control (MPC) and its variants. Section V-C reviews learning-based controllers (reinforcement learning, imitation learning, and model-based RL). Section V-D summarizes hybrid designs that combine model-based and learned components. Section V-E treats impedance and force control as a cross-cutting ingredient that appears across these families. For each family we interpret representative implementations along the four axes of model dependence, computation, safety, and adaptability introduced below.

Throughout Section V, vectors are denoted by bold lowercase symbols (e.g., $\boldsymbol{q}, \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{\tau}$) and matrices by bold uppercase symbols (e.g., $\boldsymbol{M}, \boldsymbol{J}, \boldsymbol{Q}$). Generalized coordinates are $\boldsymbol{q}$ with velocities $\dot{\boldsymbol{q}}$ and accelerations $\ddot{\boldsymbol{q}}$. Stacked contact forces are $\boldsymbol{f}_c$ (including ground reaction forces and interaction forces when applicable). The contact Jacobian is $\boldsymbol{J}_c(\boldsymbol{q})$, and the friction/unilaterality feasible set is denoted by $\mathcal{F}_\mu$ (or equivalently $\boldsymbol{A}_\mu \boldsymbol{f}_c \leq \boldsymbol{0}$ for a linearized cone).

Several alternative taxonomies could be used to organize loco-manipulation controllers. For example, prior surveys on legged locomotion and mobile manipulation group methods by planning horizon (reactive vs. predictive vs. offline planning) or by architectural decomposition (decoupled locomotion/manipulation vs. unified whole-body stacks) [17], [19], [21]. One could also classify controllers purely by mathematical tool-e.g., quadratic programming, trajectory optimization, differential dynamic programming, or deep reinforcement learning. However, these views tend to blur two questions that are central for loco-manipulation: *what* knowledge about the robot and environment is assumed (explicit models vs. data-driven approximations), and *how* that knowledge is exploited (instantaneous vs. predictive vs. combined schemes). Our model-based/optimization-based/learning-based/hybrid taxonomy is chosen to make these trade-offs explicit. Model-based controllers use analytic rigid-body/contact models in predominantly myopic ways (e.g., inverse-dynamics whole-body control), optimization-based methods extend this to finite-horizon optimal control (MPC and variants), learning-based methods rely primarily on data to represent dynamics or policies, and hybrid approaches deliberately combine these ingredients within a single stack. Importantly, these families are not mutually exclusive-for instance, WBC often contains QP layers and MPC almost always uses explicit models-but in practice they correlate strongly with different assumptions about model fidelity, data availability, and computational budget. We therefore adopt this coarse but control-relevant structure and then analyze each family along the axes of model dependence, computation, safety, and adaptability introduced below.

In this section we move beyond listing representative implementations and instead analyze these families along four recurring axes:

- Model dependence and robustness: how strongly performance relies on accurate rigid-body/contact models and state estimation.
- Computation and scalability: whether real-time performance is achievable for high-DoF quadrupeds with arms.
- Contact handling and safety: how explicitly the method enforces friction cones, joint/torque limits, and HRI safety constraints.
- Adaptability and generalization: how well the method handles changing terrain, loads, tasks, and perception errors.

TABLE 4. Overview of the proposed loco–manipulation task taxonomy, derived from a bottom-up coding of tasks in the literature along six control-relevant axes. Each row lists representative task instances, common controller families used for that setting, and example references, highlighting how changes along a single axis tend to shift the preferred control strategy.

Axis	Levels	Representative task instances	Typical controller family	Representative refs.
Interaction type	Prehensile	Pick-and-place on rough terrain; handover while standing/walking	WBC/Impedance; MPC (WB); RL+Grasp primitives	[1], [2], [51], [52]
	Non-prehensile	Door/handle push; object sliding/dragging; collision-aware body-arm motion near obstacles	MPC (WB); WBC with contact constraints	[1], [53]
Dynamics	Static / Quasi-static	Posture regulation with low base accelerations; compliant contact force tracking	WBC/Impedance; QP-based whole-body ID	[3]
	Dynamic	Manipulate while moving; momentum exchange; base reorientation during arm motion	Holistic/Hierarchical MPC; RL (unified policy); Hybrid RL+MPC	[1], [2]
Terrain	Flat	Indoor flat floors; ramps with known slope	Any (WBC/MPC/RL)	[1], [2]
	Uneven	Steps, rubble; rough outdoor surfaces	MPC with contact planning; robust WBC	[51]
	Compliant/uncertain	Soft mats, deformable ground; friction variation	Impedance/admittance; model adaptation	[3]
Payload	Light ($< 5\%$ mass)	Tool use; small object transport	Any (WBC/MPC/RL)	[1]
	Medium (5-15%)	Box carrying while walking; co-manipulation of moderate loads	Learning-based locomotion + model-based manipulation; MPC	[1], [10]
	Heavy ($> 15\%$)	Hydraulic arm handling heavy objects; large disturbance rejection	GRF optimization; stability-margin MPC/WBC	[54]
Coordination	Single robot	Solo loco-manipulation in unstructured environments	MPC/WBC; RL; Hybrids	[1], [2]
	Human-robot	Handover; collaborative lifting; posture optimization for ergonomics	Hierarchical optimization; WBC	[52]
	Teleoperation (MR/VR)	Search-and-rescue, remote inspection with shared autonomy	Teleop + WBC/MPC; shared control	[15]
	Multi-robot (rare)	Cooperative transport/manipulation with multiple platforms	Distributed planning/control (surveyed generally)	[17]
Sensing	Proprio-only	State-estimation-driven loco-manipulation; force-limited interactions	MPC/WBC with contact constraints	[1], [3]
	Proprio+vision	Visual servoing; object-aware policies; perception-in-the-loop teleop	RL (vision); Teleop MR; vision-in-the-loop MPC	[2], [15]

TABLE 5. Core symbols used in Section V.

Symbol	Meaning
$\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}$	generalized coordinates, velocity, acceleration (floating base + joints)
$\boldsymbol{\tau}$	actuated joint torques; S selects actuated joints
$\boldsymbol{M}(\boldsymbol{q}), \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}), \boldsymbol{G}(\boldsymbol{q})$	inertia, Coriolis/centrifugal, gravity terms
$\boldsymbol{J}_c(\boldsymbol{q}), \boldsymbol{f}_c$	contact Jacobian and stacked contact forces
$\boldsymbol{x}_k, \boldsymbol{u}_k$	MPC state and control input at step k
$\boldsymbol{Q}, \boldsymbol{R}, \boldsymbol{Q}_f$	MPC weighting matrices
$\mathcal{F}_\mu$	friction/unilaterality feasible set (contact constraints)

Figure 6 summarizes the forces and coordinate frames that all methods must ultimately reason about, regardless of whether they rely on an explicit model or learned representations.

A key goal of this section is not only to categorize controller families but to *critically synthesize* where they fail in practice and what mitigations are supported by reported evidence. To make this explicit, Tables 20–21 (Tables 18–19 in the final manuscript) consolidate recurring failure manifestations (e.g., slip and infeasibility under contact uncertainty, solver overruns under fast mode switching, unsafe transients under strict HRI constraints) and the corresponding mitigations reported in the cited literature (e.g., disturbance-aware prediction, impedance shaping, hierarchical decomposition, and hybrid safety envelopes). In each of the following subsections, we therefore pair the description of each control family with a short, structured "Failure modes and limitations" paragraph that points the reader to the relevant entries in Tables 20–21 and explains *why* the breakdown occurs for that family.

In addition to these four axes, each subsection explicitly comments on how the corresponding control family fares in terms of: (i) scalability to higher DoF and more complex task settings, (ii) generalization across the task taxonomy in Section IV, and (iii) the level of evidence for robust operation in long-duration, real-world deployments. Our goal is therefore not only to summarize representative implementations, but also to question the practical limits and transferability of each methodological approach.

Table 6 complements the taxonomy overview in Table 4 by rating *controller families* (rather than task levels) against the same six axes. The ratings reflect recurring empirical patterns in the cited literature: model-based WBC/impedance excels when explicit constraint handling and safe interaction dominate, MPC is strongest when prediction over contact transitions and load transfer is critical, learning-based

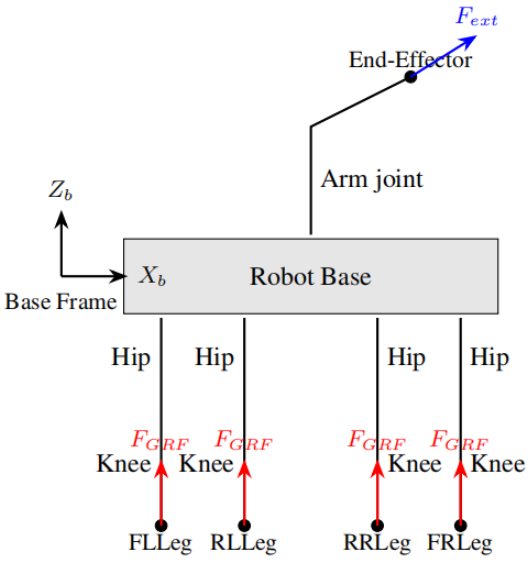


FIGURE 6. Quadruped loco-manipulation: coordinate frames, joint states, and contact/interaction forces. Any whole-body controller must distribute ground reaction forces while coordinating base and arm motion.

methods excel under broad variability and rich perception but require safety scaffolding, and hybrids increasingly dominate because they allocate hard feasibility to model-based layers and adaptation to learned components.

A. MODEL-BASED METHODS

Model-based control approaches explicitly use rigid-body and contact models to compute control actions. Typical instances include whole-body control inverse dynamics (ID-WBC) with (hierarchical) quadratic programs that track task accelerations while enforcing torque, joint, and friction-cone constraints, as well as operational-space / centroidal formulations for balancing with an arm [26], [27], [35], [62].

The main appeal of these methods is that they make the structure of the robot-floating-base dynamics, multi-contact constraints, and base-arm coupling-explicit. This yields: (i) interpretable task priorities (e.g., keep balance before tracking end-effector pose), (ii) strong constraint handling, and (iii) compatibility with safety standards where guarantees on joint, torque, and friction limits are essential. For tasks such as precise object handover or force-controlled door opening on relatively structured terrain, these properties are difficult to match with pure learning-based approaches.

A standard floating-base rigid-body model with contacts is [34]:

$$\boldsymbol{M}(\boldsymbol{q})\,\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\,\dot{\boldsymbol{q}} + \boldsymbol{G}(\boldsymbol{q}) = S^{\top}\boldsymbol{\tau} + J^{\top}(\boldsymbol{q})\,\boldsymbol{F}_{\text{ext}}, \tag{24}$$

where $\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}$ are generalized coordinates, velocities, and accelerations; $\boldsymbol{M}, \boldsymbol{C}, \boldsymbol{G}$ are inertia, Coriolis, and gravity; S selects actuated joints; $\boldsymbol{\tau}$ are joint torques; and $\boldsymbol{F}_{\text{ext}}$ collects external/contact forces. Model-based controllers differ mainly in how they choose $\ddot{\boldsymbol{q}}$ and $\boldsymbol{F}_{\text{ext}}$, and how tightly they enforce feasibility.

Algorithm 2 Classical Whole-Body Control (WBC) With Priorities [26], [63]

Require: Joint state $(\boldsymbol{q}, \dot{\boldsymbol{q}})$, prioritized tasks $\{(J_i, \ddot{\boldsymbol{x}}_i^{\text{des}})\}$, robot model $(\boldsymbol{M}, \boldsymbol{C}, \boldsymbol{G}, S)$
Ensure: Joint torques $\boldsymbol{\tau}$
1: Update $\boldsymbol{M}(\boldsymbol{q})$, $\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})$, $\boldsymbol{G}(\boldsymbol{q})$, $J_i(\boldsymbol{q})$, $J_c(\boldsymbol{q})$
2: Form QP

$$\min_{\ddot{\boldsymbol{q}},\boldsymbol{\tau},\boldsymbol{f}_c} \sum_i w_i \left\| J_i\ddot{\boldsymbol{q}} + \dot{J}_i\dot{\boldsymbol{q}} - \ddot{\boldsymbol{x}}_i^{\text{des}} \right\|_2^2 + \lambda\|\boldsymbol{\tau}\|_2^2$$

3: **s.t.** $\boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{C}\dot{\boldsymbol{q}} + \boldsymbol{G} = S^{\top}\boldsymbol{\tau} + J_c^{\top}\boldsymbol{f}_c$, $\mathcal{A}_{\mu}\boldsymbol{f}_c \le \boldsymbol{0}$, $\underline{\boldsymbol{\tau}} \le \boldsymbol{\tau} \le \overline{\boldsymbol{\tau}}$
4: Solve for $(\ddot{\boldsymbol{q}}^{\star}, \boldsymbol{\tau}^{\star}, \boldsymbol{f}_c^{\star})$ and apply $\boldsymbol{\tau}^{\star}$

1) CLASSICAL WHOLE-BODY CONTROL (WBC)

A foundational approach is task-oriented (operational-space) WBC, which coordinates multiple primitive tasks while respecting physical constraints [62], [63]. Modern realizations employ single-level or hierarchical QPs to assign strict or soft priorities across tasks (e.g., CoM/base regulation, swing-foot/hand motion, posture) while enforcing dynamics and contact feasibility [26], [27]. The condensed HQP routine is shown in Algorithm 2.

Empirically, WBC works best when the contact schedule is known or slowly changing and when model errors (e.g., inertial parameters, friction coefficients) can be calibrated. Under those conditions, works such as [52], [55], [64] report precise whole-body tracking and safe interaction.

However, the same studies also highlight three limitations: (i) sensitivity to modeling errors and latency in state estimation, particularly on rough terrain, (ii) considerable engineering effort to tune task weights and gains across operating conditions, and (iii) challenges in scaling to very dynamic behaviors where contact switches happen faster than the QP can be reliably solved on embedded hardware.

2) INVERSE-DYNAMICS-BASED WBC

Inverse-dynamics WBC computes the torques required to realize task-space accelerations while accounting for base-arm coupling and contact feasibility. For a task $\boldsymbol{x}_t = f_t(\boldsymbol{q})$ with Jacobian J_t, a standard PD+feedforward target is

$$\ddot{\boldsymbol{x}}_t^{\star} = \ddot{\boldsymbol{x}}_d + K_p(\boldsymbol{x}_d - \boldsymbol{x}_t) + K_d(\dot{\boldsymbol{x}}_d - \dot{\boldsymbol{x}}_t), \tag{25}$$

with a consistent joint acceleration target

$$\ddot{\boldsymbol{q}}^{\star} = J_t^{\dagger}\left(\ddot{\boldsymbol{x}}_t^{\star} - \dot{J}_t\,\dot{\boldsymbol{q}}\right). \tag{26}$$

Torques and contact forces are then obtained from an inverse-dynamics QP:

$$\min_{\boldsymbol{\tau},\ddot{\boldsymbol{q}},\boldsymbol{f}_c} \quad \left\| J_t\ddot{\boldsymbol{q}} + \dot{J}_t\dot{\boldsymbol{q}} - \ddot{\boldsymbol{x}}_t^{\star} \right\|_2^2 + \lambda\|\boldsymbol{\tau}\|_2^2 \tag{27}$$

$$\text{s.t. } \boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{C}\dot{\boldsymbol{q}} + \boldsymbol{G} = S^{\top}\boldsymbol{\tau} + J_c^{\top}\boldsymbol{f}_c,\ \boldsymbol{f}_c \in \mathcal{F}_{\mu},\ \underline{\boldsymbol{\tau}} \le \boldsymbol{\tau} \le \overline{\boldsymbol{\tau}}. \tag{28}$$

Figure 7 shows a typical pipeline of the WBC.

TABLE 6. Cross-family synthesis: rating of control families against the six task-taxonomy axes (Table 4). Ratings reflect typical *best-fit regimes* and dominant bottlenecks reported across representative works in Section V (✓ strong fit, ● viable with caveats, ✗ generally weak).

Control family	Interaction	Dynamics	Terrain	Payload	Coordination	Sensing	Typical bottlenecks / notes (evidence)
Model-based (ID-WBC / QP + impedance)	✓ (preh.) ● (non-preh.)	●	●	● (✓ if calibrated)	●	● (proprio) ✗ (vision-heavy)	Strong explicit constraints and safe interaction via impedance/QP, but sensitive to model/contact mismatch and requires tuning; works well on structured terrain and moderate dynamics [26], [27], [52], [55].
Optimization-based (MPC / WB-MPC / hierarchical MPC)	● (preh.) ✓ (non-preh.)	✓	✓	✓	●	● (proprio) ● (vision-in-loop)	Best for dynamic stability, contact scheduling, and load transfer with explicit constraints; limited by solver rate/model fidelity; large WB-MPC can be compute-hungry [1], [28], [56], [57].
Learning-based (RL/IL/MBRL)	● (preh.) ● (non-preh.)	✓	●	●	●	✓ (vision+proprio)	High adaptability and perception integration; weakest on explicit safety/certification without additional structure; sensitive to distribution shift beyond training coverage [2], [58], [59].
Hybrid (learning + MPC/WBC/residuals/safety filters)	✓	✓	✓	✓	✓	✓	Combines explicit constraint handling with data-driven adaptation; integration complexity and interface tuning are major practical challenges; increasingly dominant in recent work [5], [10], [60], [61].

Compared to purely kinematic control, ID-WBC can explicitly trade off torque effort vs. tracking accuracy, and can shape compliance in task space by embedding impedance objectives in the QP [20], [55]. This is crucial for HRI scenarios (handover, co-carrying) where transparent interaction and safety are non-negotiable.

On the downside, several works report that the tight coupling between model fidelity and control performance can make ID-WBC fragile when contact locations or surface properties are poorly known [27], [52], [65]. In practice, this often leads to ad hoc modifications (friction margins, contact blending, saturations) which partly erode the theoretical guarantees.

3) EXPLICIT CONTACT CONSTRAINTS

For completeness, unilateral/friction/no-slip constraints used by WBC and MPC can be written as

$$\mathcal{A}_\mu \boldsymbol{f}_c \leq \mathbf{0}, \; J_c(\boldsymbol{q})\ddot{\boldsymbol{q}} + \dot{J}_c(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} = \mathbf{0}, \tag{29}$$

with $\mathcal{A}_\mu$ defining linearized friction cones and unilaterality [26], [27]. All model-based controllers must in effect respect these constraints; the difference lies in whether they are enforced explicitly (via inequalities) or only implicitly through heuristics and saturation.

4) ASSESSMENT OF MODEL-BASED METHODS

Table 8 summarizes representative model-based loco-manipulation controllers. Rather than listing platforms, we emphasize how each work navigates the trade-offs highlighted above.

Overall, model-based methods are most compelling when constraints and safety dominate and when sufficient modeling and calibration effort is acceptable. Their limitations become pronounced in rapidly varying environments, under large model uncertainty, or when compute is severely constrained.

From a critical standpoint, current model-based loco-manipulation stacks also exhibit limited cross-platform generalization: almost all reported systems are tuned to a single robot, a specific arm configuration, and a narrow task set, with substantial manual effort invested in model calibration and gain/priority tuning. There is little evidence that a given WBC design can be ported between quadruped platforms or terrain regimes without significant retuning, and long-term field deployments that confront wear, sensor degradation, or evolving payloads are largely absent. Thus, while model-based approaches are strong in local interpretability and constraint handling, their scalability and ''out-of-the-box'' practicality across robots and applications remain open issues.

Across the surveyed corpus, the dominant breakdown mechanisms of model-based WBC/ID-QP stacks are sensitivity to contact/model mismatch and estimator latency, which manifest as slip, conservative saturation, or degraded force/pose tracking under friction variation or uncertain footholds (Table 20). These limitations are commonly addressed by adding impedance shaping and robustness margins (contact blending, friction margins), but the resulting performance depends strongly on tuning and calibration effort (Table 20). This evidence supports our broader conclusion that model-based WBC excels under structured contact

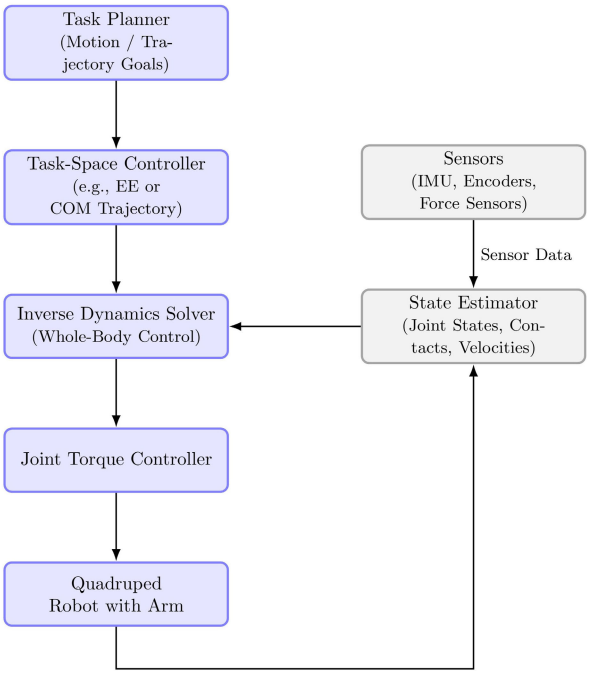


FIGURE 7. Architecture of inverse-dynamics WBC for a quadruped with a manipulator: task planner → task-space controller → inverse-dynamics QP → torque execution with state estimation, following [27], [52].

regimes and strict constraint enforcement, but becomes brittle when contact uncertainty and fast mode transitions dominate.

These limitations, particularly the difficulty of anticipating fast contact changes and arm-induced disturbances, motivate predictive formulations such as MPC, which we discuss next in Section V-B.

B. OPTIMIZATION-BASED METHODS: MPC AND VARIANTS

Model Predictive Control (MPC) is widely favored for loco-manipulation because it optimizes over a finite horizon while naturally handling floating-base dynamics, multi-contact constraints, and multi-objective trade-offs. Conceptually, MPC extends model-based control from a *myopic* (instantaneous) view to a *predictive* one: it can explicitly reason about future contact switches, arm-induced disturbances, and task goals [10], [28], [29], [66].

The canonical MPC stack is shown in Figure 8 and Algorithm 3. Over a horizon N, a standard MPC formulation minimizes

$$\min_{\mathbf{u}_{0:N-1}} \sum_{k=0}^{N-1} \left\|\mathbf{x}_k - \mathbf{x}_k^{\text{ref}}\right\|_{\mathbf{Q}}^2 + \left\|\mathbf{u}_k\right\|_{\mathbf{R}}^2 + \left\|\mathbf{x}_N - \mathbf{x}_N^{\text{ref}}\right\|_{\mathbf{Q}_f}^2 \quad (30)$$

subject to discrete dynamics and constraints:

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k),\ \mathbf{x}_{\min} \le \mathbf{x}_k \le \mathbf{x}_{\max},$$
$$\mathbf{u}_{\min} \le \mathbf{u}_k \le \mathbf{u}_{\max}, \quad (31a)$$
$$\mathbf{f}_c \in \mathcal{F}_{\text{friction}}, \quad J_{ee}(q)\,\ddot{q} + \dot{J}_{ee}(q,\dot{q}) = \ddot{\mathbf{x}}_{ee}. \quad (31b)$$

TABLE 7. When to prefer ID-WBC/impedance for loco-manipulation (✓ best fit, ● viable with caveats).

Condition	Fit	Justification (evidence)	Notes / representative refs.
Prehensile tasks on flat or mildly uneven terrain	✓	Constraint-consistent torque/GRF allocation and task-space compliance are primary requirements; calibrated rigid-body/contact models typically suffice.	Accurate force/pose control with impedance shaping [20], [52].
Moderately dynamic interaction while walking	●	Myopic QP/WBC can succeed when contact schedule changes slowly; predictive layers are beneficial when coupling and timing dominate (cf. Table 19).	Works in practice; hybrids/MPC often improve transients [27], [52].
Strict safety / explicit limits (HRI, torque/contacts)	✓	Hard inequality constraints (torque, friction cones, unilateral contact) and impedance/passivity structure are directly enforceable in HQP/ID-QP (see Table 20).	QP constraints and whole-body impedance for safe interaction [20], [26].
High model uncertainty / fast terrain change	●	Performance becomes sensitive to friction/contact mismatch and estimator latency; robustness margins or adaptive/disturbance-aware layers are typically required (Tables 20–21).	Sensitive; add MPC/learning residuals and impedance envelopes [27], [61], [65].

Compared to WBC, MPC trades increased computational load for the ability to *anticipate* base-arm interactions and contact transitions. This makes it attractive for tasks where the timing of contacts and load transfer is critical (e.g., trotting while carrying an object).

Beyond MPC, *trajectory optimization (TO)* solves full motion plans intermittently or offline; *quadratic programming (QP)* layers appear inside whole-body inverse-dynamics controllers; *nonlinear programming (NLP)* addresses high-DoF constrained planning; and *sequential convex programming (SCP)* handles nonconvexities in contact-rich settings. These methods mainly differ in how often they are solved and how strongly they couple to low-level tracking.

1) WHOLE-BODY MPC (WB-MPC)

WB-MPC jointly optimizes locomotion and manipulation by embedding centroidal/floating-base dynamics, contact constraints, and arm kinematics in a single OCP. Dadiotis et al. [28] demonstrate real-time WB-MPC on

TABLE 8. Representative model-based loco-manipulation controllers (model-only subset) and their practical trade-offs.

Reference (Year)	Method (Family)	Platform	Notes (strengths / limits)
Bellicoso et al. (2019) [52]	Hierarchical WBC (ID-QP)	ANYmal + arm (ALMA)	Collaborative loco-manipulation; manipulability and posture optimization; relies on calibrated models and careful priority tuning
Fahmi et al. (2019) [65]	"Passive" WBC (QP)	HyQ	Robust dynamic locomotion with passivity-inspired design; performance degrades with large modeling errors or fast terrain changes
Li et al. (2022) [64]	ID-WBC (QP)	Aliengo + 5-DoF arm	Accurate tracking of base-arm coupling; optimization time becomes critical for highly dynamic tasks
Risiglione et al. (2022) [55]	Cartesian impedance in QP	HyQ + 7-DoF KUKA	Safe compliance and transparent interaction; impedance design is hardware- and task-specific
Sleiman et al. (2021) [1]	Whole-body MPC (borderline between model-based and optimization-based)	ANYmal-C + arm	Predictive contact and arm coordination; exposes computational limits of unified formulations

CENTAURO (37 DoF, dual-arm), interfacing directly to joint impedance without a separate WBC layer. Related work on representation-free MPC [29] and high-speed locomotion [66] shows that predictive formulations can achieve highly dynamic maneuvers.

Trade-offs: WB-MPC offers the most principled way to coordinate base and arm under tight safety limits, but exposes the full computational burden of solving a large nonlinear OCP at high rates. In practice, this leads to compromises on horizon length, model complexity (e.g., centroidal vs full dynamics), or solver accuracy. When compute is limited or when perception introduces high-latency feedback, WB-MPC alone may not be sufficient.

2) DISTURBANCE-COMPENSATION MPC (DC-MPC)

When the manipulator induces significant base disturbances or the robot handles unknown loads, DC-MPC augments prediction with disturbance models/estimators. Wang et al. [56] show that modeling arm-induced wrenches improves stability during trotting+manipulation. Related schemes combine disturbance-aware low-level controllers with high-level estimators [67], [68].

Here the key insight is *why* compensation matters: standard MPC tends to underestimate the effect of unmodeled wrenches, leading to base drift and loss of contact feasibility. DC-MPC mitigates this at the price of additional model complexity and estimator design, making it preferable when payloads and interaction forces are large compared to the robot weight.

3) HIERARCHICAL MPC (H-MPC)

H-MPC decomposes the problem across timescales or abstraction levels (e.g., contact timing vs whole-body motion). Rigo et al. [57] use a two-level MPC for non-prehensile pushing, while Sombolestan and Nguyen [69] extend this idea to adaptive cooperative manipulation. Khan et al. [70] apply hierarchical optimum control with an MPC layer to wheel-legged systems.

The main advantage is tractability: high-level plans can be computed with longer horizons and reduced dynamics, whereas the low-level MPC focuses on short-horizon stabilization. The main limitation is cross-layer consistency. If high-level assumptions (e.g., feasible contact sequences) do not match low-level realities (e.g., slipping or perception errors), the hierarchy can fail in ways that are harder to debug than a monolithic controller.

Algorithm 3 MPC-Based Loco-Manipulation Control Loop

Require: initial state $\mathbf{x}_0$, reference $\mathbf{x}^{\text{ref}}_{0:N}$, model f, weights $(\mathbf{Q}, \mathbf{R}, \mathbf{Q}_f)$

1: **while** robot is active **do**
2: Measure $\mathbf{x}_k$; update contact schedule and task references
3: Formulate constrained OCP over horizon N
4: Solve for $\mathbf{u}^{\star}_{0:N-1}$ and apply $\mathbf{u}^{\star}_0$
5: Shift horizon; $k \leftarrow k+1$

4) DATA-DRIVEN AND LEARNING-INFUSED MPC (DL-MPC)

DL-MPC reduces reliance on exact models by learning dynamics or calibrating MPC parameters from data. Examples include combining a learned locomotion policy with model-based manipulation [10], sample-efficient dynamics learning for loco-manipulation [71], and adaptive or nonlinear MPC customized for quadrupeds with non-negligible arm dynamics [72], [73].

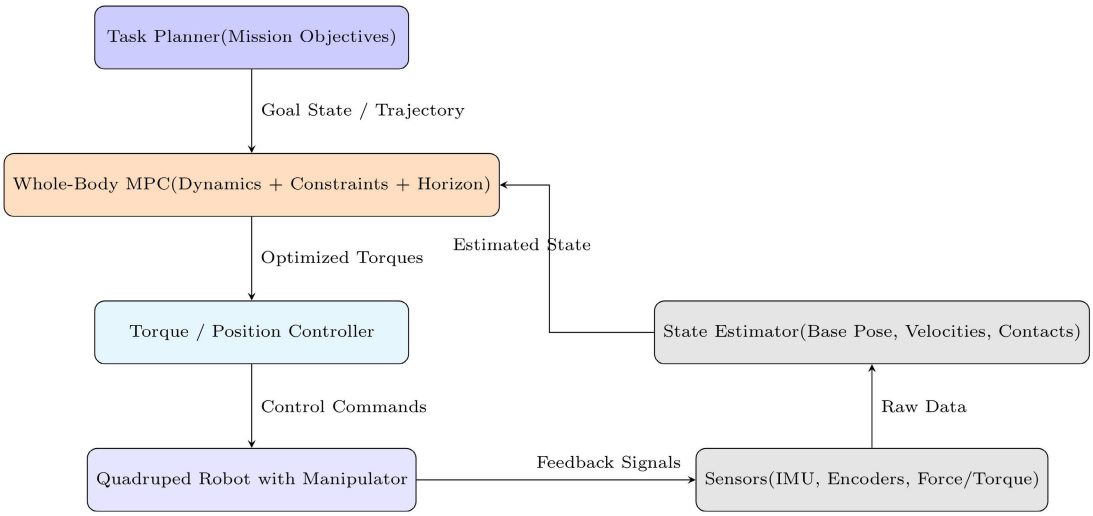


FIGURE 8. Canonical MPC-based loco-manipulation stack. High-level task references feed a whole-body MPC with centroidal/floating-base dynamics and contact constraints; a low-level torque/impedance controller executes the first control input at each cycle while state estimation closes the loop.

These methods are attractive when model errors are the dominant failure mode of classical MPC. However, they also inherit the data requirements and sim-to-real challenges of learning-based approaches while retaining the computational cost of solving OCPs.

5) ASSESSMENT OF OPTIMIZATION-BASED METHODS

Table 9 compares representative MPC strategies, and Table 10 summarizes ''when to use what''. The overall picture is that MPC is most compelling when predictive reasoning about contact and load transfer is crucial and sufficient compute is available. When compute or modeling resources are limited, hierarchical or hybrid approaches become more attractive than monolithic MPC.

At the same time, both WBC and MPC remain fundamentally model-centric and can be brittle under severe model mismatch or perception errors. This motivates learning-based approaches, which we review in Section V-C.

Optimization-based methods most often fail via computational overload (missed deadlines/solver overruns) or infeasibility under fast contact switching and high uncertainty, particularly when the online model is pushed toward higher fidelity (Table 21). Reported mitigations include hierarchical decomposition (separating long-horizon planning from short-horizon stabilization), reduced-order predictive layers, and disturbance-aware formulations for arm-induced wrenches (Tables 20–21). This clarifies the practical trade-off behind MPC: predictive contact and load-transfer reasoning is strongest when compute and state estimation quality support consistent real-time solves.

Critically, most whole-body MPC demonstrations reported to date are confined to well-instrumented laboratory settings with carefully engineered state estimators and terrain models. Reported solve times and hardware limitations suggest that scaling these formulations to larger fleets, richer perception stacks, or severely resource-constrained onboard computers will require either substantial algorithmic advances or aggressive model reduction. Moreover, there is little evidence that a single MPC configuration can be deployed unchanged across different quadruped platforms or substantially different loco-manipulation tasks. This tempers expectations about the near-term generalizability and real-world practicality of monolithic WB-MPC, and motivates the hierarchical and hybrid variants discussed in this section.

C. LEARNING-BASED ALGORITHMS

Learning-based control (reinforcement learning, imitation learning, and data-driven MPC) has enabled robust loco-manipulation in unstructured environments by mitigating dependence on exact models [2], [9], [59], [74]. Rather than only summarizing architectures, we focus on why these methods succeed where model-based approaches struggle and what new limitations they introduce. A high-level stack is shown in Figure 9, with runtime listed in Algorithm 4. Representative methods are compared in Tables 11-13.

The main motivation is to handle phenomena that are hard to model: complex contact friction, actuator backlash, perception errors, and human behavior in HRI.

Policies π_θ map observations $\boldsymbol{o}_t$ (proprioception, contacts, vision) to actions $\boldsymbol{a}_t$:

$$\boldsymbol{a}_t = \pi_\theta(\boldsymbol{o}_t), \tag{32}$$

trained either by maximizing a task reward (RL),

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T} \gamma^t \, r(\boldsymbol{o}_t, \boldsymbol{a}_t)\right], \tag{33}$$

TABLE 9. Representative MPC strategies for quadruped loco-manipulation, emphasizing key advantages and failure modes.

Reference (Year)	MPC Strategy	Robot / Config.	Advantages	Limitations
Dadiotis et al. (2023) [28]	Whole-Body MPC (centroidal + kinematics; DDP/OCS2)	CENTAURO (37-DoF dual-arm quadruped)	Real-time whole-body coordination; no separate WBC layer	High compute and strong model dependence; limited by solver rate
Wang et al. (2023) [56]	Compensation-Based MPC (arm-induced wrench modeling)	Custom quadruped + arm	Improved stability during trotting+manipulation	Requires disturbance models/estimators; tuning effort
Rigo et al. [57]	Hierarchical MPC for non-prehensile pushing	Quadruped manipulator	Optimizes contact placement/forces and robot motion separately	Cross-layer tuning and consistency; compute for contact search
Sombolestan & Nguyen [69]	Hierarchical adaptive loco-manipulation (with MPC layer)	Multi-robot cooperative manipulation	Robust to unknown payloads; decentralized coordination	Multi-agent complexity and estimation quality
Ding et al. (2020) [29]	Representation-free MPC (rotation-matrix dynamics)	Quadruped (locomotion)	High-rate MPC; handles 3D orientation and dynamic maneuvers	Locomotion-focused; extension to arm coupling is nontrivial
Khan et al. (2024) [70]	Hierarchical optimum control + MPC	Pegasus (wheel-legged quadruped)	Hybrid locomotion planning/control; efficient stages	Requires precise modeling and more complex software stack
Ma et al. (RAL'22) [10]	Learning-infused: RL base + model-based arm (uses MPC wrench prediction)	ANYmal + arm (sim/real)	Disturbance-aware base; good sim-to-real for mobile manipulation	Needs a trained locomotion policy and integration effort
Turrisi et al. (2024) [73]	MPC with *passive* arm for cooperative carrying (PACC)	Quadruped + passive arm	Low energy and simple control for carrying; robust co-manipulation	Limited dexterity and task specificity
Nonlinear MPC (2025) [72]	NMPC tailored for arm+quadruped coupled dynamics	Quadruped + manipulator	Captures strong arm/base coupling; efficient letter-scale solver	Early-stage; hardware validation and robustness still open

or by matching expert demonstrations (IL),

$$\mathcal{L}(\theta) = \sum_{t=0}^{T} \left\| \pi_\theta(\boldsymbol{o}_t) - \boldsymbol{a}_t^{\exp} \right\|_2^2 . \tag{34}$$

Learning-based controllers often achieve behaviors that are difficult to engineer by hand (e.g., agile leg-based manipulation [75] or vision-conditioned object retrieval [58]). However, their performance is tied to the coverage of the training data and they typically offer weaker guarantees on safety and contact feasibility than WBC or MPC.

Recent hierarchical RL studies in constrained environments illustrate a broader trend toward *structure in the policy* (task decomposition and constraint-aware decision layers), which is directly relevant for loco-manipulation where contact and feasibility constraints must be satisfied under distribution shift [76].

1) MODEL-BASED RL (MBRL)

MBRL learns predictive models and uses them for planning/control, improving sample efficiency and enabling look-ahead under contact changes [71], [77]. Compared to classical MPC, the learned model allows adaptation to unmodeled dynamics (e.g., wear, payload changes), but also introduces model bias: errors in the learned model can mislead the planner in systematic ways. This trade-off is attractive when data collection is feasible and when the environment varies faster than a hand-crafted model can be maintained.

2) MODEL-FREE RL (MFRL)

MFRL learns policies directly from interaction [74], [75]. These methods have shown impressive adaptability (e.g., leg-only manipulation without an arm), but require large-scale data and strong safety mechanisms to avoid catastrophic failures during exploration. For quadruped loco-manipulation, this often translates into extensive simulation training with domain randomization and cautious sim-to-real transfer.

3) IMITATION LEARNING (IL)

IL leverages demonstrations to bypass reward engineering and can provide fast deployment when high-quality demonstrations exist [58]. Its main weakness is distribution shift: the policy may encounter states not covered by the demonstrations, leading to cascading errors. DAgger-like aggregation or IL+RL fine-tuning alleviates but does not fully eliminate this issue.

4) HYBRID LEARNING+CONTROL

Hybrid learning+control (discussed further in Sec. V-D) mixes learned policies with structured controllers, often to combine adaptability with safety [5], [10], [60], [61]. From a control-design perspective, these hybrids are best seen as

TABLE 10. "When to use what" for optimization-based loco-manipulation (✓ best fit, ● viable with caveats, ✗ not ideal).

Condition / Scenario	WB-MPC	DC-MPC	H-MPC	DL-MPC	Passive-arm + MPC	Justification (evidence)
Base-arm coupling and contact feasibility must be co-optimized under tight safety limits	✓	●	●	●	✗	Unified OCP captures coupling and explicit constraints; demonstrated whole-body MPC at large system scale (e.g., 37-DoF) and unified quadruped+arm MPC with reported update-rate anchors (Table 19) [1], [28].
Significant, partly predictable disturbances from the arm or environment	●	✓	●	●	●	Disturbance-aware prediction/estimation reduces base drift and constraint violations caused by arm-induced wrenches; supported by compensation-based MPC and hybrid MPC+learning evidence (Tables 20–21) [5], [56], [73].
Compute limits: need separation between long-horizon planning and short-horizon stabilization	●	●	✓	●	●	Hierarchical decomposition reduces online NLP burden and improves deadline reliability; aligns with solver-overrun failure modes and mitigations summarized in Table 21 [29], [57], [69].
Model fidelity insufficient or operating envelope is broad/uncertain	●	●	●	✓	✗	Learning-infused / data-driven MPC improves robustness to mismatch while retaining constraint handling; trade-off is data/validation burden (Table 21) [10], [61], [71].
Collaborative carrying with modest dexterity needs	●	●	●	✗	✓	Passive/compliant arm reduces disturbance injection and simplifies contact consistency, improving robustness when dexterity demands are low [73].

Algorithm 4 Learning-Based Loco-Manipulation Controller (runtime)

Require: trained policy $\pi_\theta(\cdot)$

while control loop **do**
  acquire observation $\boldsymbol{o}_t$ (joint states, base pose, contacts, vision)
  $\boldsymbol{a}_t \leftarrow \pi_\theta(\boldsymbol{o}_t)$
  if $\boldsymbol{a}_t$ is not torque **then**
    $\boldsymbol{\tau}_t \leftarrow f_{\text{control}}(\boldsymbol{a}_t, \boldsymbol{q}_t, \dot{\boldsymbol{q}}_t)$ ▷ e.g., inverse dynamics/impedance
  else
    $\boldsymbol{\tau}_t \leftarrow \boldsymbol{a}_t$
  send $\boldsymbol{\tau}_t$ to actuators

targeted responses to the failure modes of pure RL (lack of guarantees) and pure MPC/WBC (lack of robustness to modeling errors).

5) ASSESSMENT OF LEARNING-BASED METHODS

Tables 11-13 summarize representative methods and qualitatively rate their generalization, adaptability, and data efficiency. A recurring pattern is that learning excels in broad, partially observed environments but struggles to meet strict safety and certification requirements without additional structure.

Because learning-based controllers trade model dependence for data dependence, and model-based controllers trade data requirements for analytic structure and guarantees, their strengths are complementary. This complementarity explains the rise of hybrid architectures, which we survey in Section V-D.

Learning-based policies most commonly break under distribution shift beyond training coverage (new terrains/objects/lighting, rare contact states), and may violate safety/feasibility constraints without additional structure (Table 21). Mitigations reported in the literature rely on training-time coverage (domain randomization/curricula, demonstration aggregation) and deployment-time safety scaffolding (constraint layers, impedance envelopes, or model-based tracking) (Table 21). This supports our assessment that learning is strongest for adaptability and perception integration, but requires explicit feasibility/safety mechanisms for reliable loco-manipulation.

From the perspective of scalability and practicality, the main bottlenecks for learning-based loco-manipulation are data and safety. Many of the most impressive results rely on millions of simulated episodes, elaborate reset and monitoring infrastructure, or carefully staged real-world data collection campaigns. Only a small number of works report long-duration deployments in messy, partially observed outdoor environments, and almost none demonstrate "drop-in" transfer of a trained policy between different quadruped platforms or arm configurations. This gap between controlled demonstrations and everyday deployment limits the current real-world impact of pure learning-based approaches, even as they show clear potential in terms of adaptability.

D. HYBRID CONTROLLERS

Hybrid controllers combine structure from model-based optimization with learned elements, e.g., MPC or inverse-dynamics

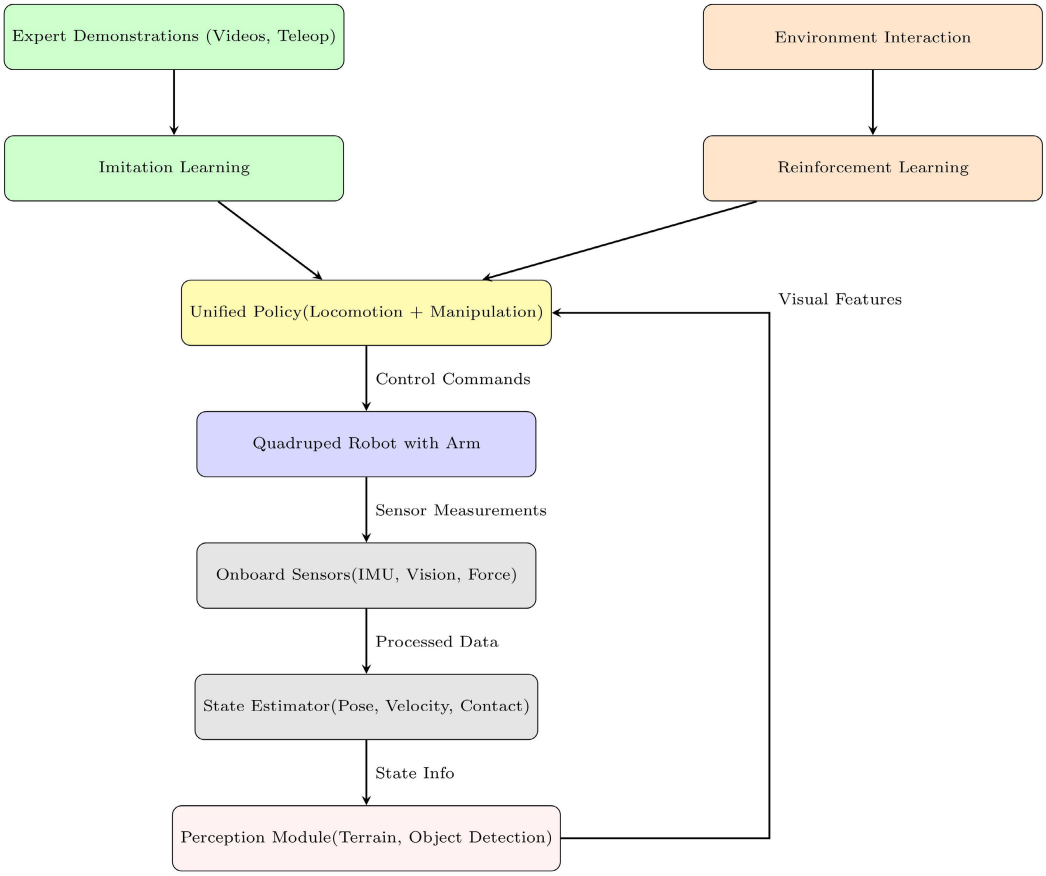


FIGURE 9. Learning-based control stack for quadruped loco-manipulation. IL/RL produce a unified whole-body policy. Perception and state estimation close the loop to adapt behavior online.

WBC for feasibility and safety with learned residuals/adaptations, or learned high-level policies supervising model-based low-level control [5], [10], [60], [61]. Typical patterns and examples are summarized in Table 14; usage guidance appears in Table 16.

A prominent recent direction is *RL-augmented model-based control*, where an optimization backbone (e.g., MPC/WBC with explicit constraints) ensures feasibility and safety, while learning provides residual adaptation or cost shaping to compensate model mismatch and contact uncertainty [61]. Such designs partially bridge the traditional separation between optimization-based and learning-based families by keeping hard constraints explicit while using data-driven modules to improve robustness in contact-rich regimes.

In practice, neither pure optimization nor pure learning fully satisfies the requirements of quadruped loco-manipulation: model-based methods struggle with unmodeled phenomena and perception errors, while learning-based methods struggle with hard safety constraints and certification. Hybrid designs explicitly allocate responsibilities: models enforce *hard* constraints (joint limits, friction cones, collision avoidance), while learned components handle *soft* aspects (cost shaping, high-level decisions, residual compensation).

Typical patterns include: (i) *planner + learned locomotion*: a learned base policy provides robust locomotion while a model-based arm optimizer ensures precise manipulation [10]; (ii) *learned residuals* on top of WBC or MPC [61]; (iii) *hierarchical hybrids* in which a learned high-level policy selects subgoals or contact modes for a lower-level WBC/MPC [60]; and (iv) *learning in the loop* where dynamics or cost terms in MPC are learned from data [5].

The main open questions are how to guarantee stability while learning is ongoing and how to manage the computational budget when solving OCPs and running neural policies concurrently.

In most hybrid stacks, a low-level impedance or force-control layer acts as the common interface between high-level planners or policies and the hardware. We therefore make impedance/force control explicit in the next subsection (Section V-E) and discuss its role as a unifying ingredient across control families.

Hybrid approaches reduce the dominant failure modes of pure families by allocating hard feasibility/safety to model-based layers and adaptation to learned components;

TABLE 11. Representative learning-based loco-manipulation controllers for quadruped robots with manipulators.

Reference (Year)	Strategy	Robot / Config.	Advantages	Limitations
Hoffman et al. (2025) [71]	MBRL with Bayesian NNs (physics priors)	Spot + arm	Sample-efficient model learning; accurate tracking	Model bias; nontrivial training/priors
Cheng et al. (2025) [61]	RL-augmented optimal control (RAMBO)	Quadruped + arm	Robust to unmodeled effects; balanced planning/reactivity	Integration complexity; compute
Ferrolho et al. (2023) [5]	RL locomotion + MPC manipulation (RoLoMa)	Quadruped + arm	Handles arm-induced wrenches; coordinated whole-body	Disturbance modeling; solver overhead
Qiu et al. (2024) [58]	Whole-body IL with vision-language prompts	Quadruped + arm	Generalizable long-horizon skills	Demo quality/coverage; perception dependence
He et al. (2024) [76]	Hierarchical RL + BC (leg-only manipulation)	Quadruped (no arm)	Reduces hardware; agile leg manipulation	Limited dexterity vs. dedicated arms
Portela et al. (2024) [59]	RL-based whole-body *force* control	Quadruped + arm	Compliant interaction; intuitive teleop	Force estimation/safety constraints
Jiang et al. (2024) [78]	RL with Reward Fusion Module	Wheeled-quadruped + arm	Precise 6D EE tracking; smooth transitions	Reward shaping; stability in training
Zeng et al. (2023) [79]	Data-driven adaptive MPC (error model)	Quadruped + arm	Compensates model errors; robustness	Error-model training; compute
Kovalev et al. (2023) [60]	Predictive RL + MPC (hybrid)	Quadruped + arm	Stability via planning + learning	Tuning/handshake between modules
Arena et al. (2023) [80]	Data-driven MPC	Mini Cheetah + arm	Better trajectory accuracy; slippery surfaces	Terrain/task specificity; generalization
Yao et al. (2023) [67]	Disturbance-predictive control (learning-aided)	Quadruped + arm	Robust to manipulator wrenches	Estimation quality; complexity

TABLE 12. Assessment of learning-based controllers along generalization, adaptability, and data efficiency.

Control family	Generalization	Adaptability	Data efficiency	Representative refs.
Model-Based RL (MBRL)	High	High	Moderate-High	[71], [78]
Model-Free RL (MFRL)	Moderate	High	Low	[74], [76]
Imitation Learning (IL)	Moderate-High	Moderate	High	[58], [76]
Hybrid (RL + MPC / Optimal Control)	High	High	Moderate	[5], [60], [61]
Data-Driven / Adaptive MPC	Moderate	High	Moderate-High	[67], [79], [80]
RL-based Whole-Body *Force* Control	Moderate	Moderate-High	Moderate-Low	[59]
Vision-Conditioned IL/RL (VBC)	High	High	Moderate-Low	[58], [75]

however, they introduce integration failures (mismatched interfaces, mode-transition brittleness) and increased software/verification complexity (Table 21). The literature therefore suggests that the primary bottleneck for hybrids is no longer feasibility *in principle*, but maintainable and verifiable system integration under compute and perception constraints.

While hybrid stacks are arguably the most promising direction in terms of combining safety with adaptability, they also raise new challenges for scalability and maintainability. Composing learned components with optimization layers increases software complexity, complicates debugging, and makes it harder to provide end-to-end guarantees when any of the learned modules are retrained or updated. At present, most hybrid loco-manipulation systems are proof-of-concept prototypes evaluated on a single platform; systematic studies of how such architectures behave under hardware changes, large task shifts, or prolonged deployment are largely missing.

E. IMPEDANCE AND FORCE CONTROL

Impedance/force-control layers provide real-time compliance and contact robustness and are frequently embedded inside model-based QPs (as task-space impedance objectives) or used to post-shape learned actions for safety [20], [55], [59], [62]. The article treat them as a cross-cutting ingredient that appears in model-based, optimization-based, and hybrid stacks.

1) ROLE IN LOCO-MANIPULATION AND HRI

For collaborative tasks such as human-robot handover or co-transport, impedance enables safe and transparent exchanges of momentum and force. Quadruped platforms with arms exploit posture and impedance modulation to keep the base stable while accommodating human motion [52], [55]. Learned whole-body force control further improves natural interaction but typically relies on model-based constraints or saturation to guarantee safety [59]. Selected approaches

TABLE 13. When to use what: **Learning families** (complements Table 12).

Condition / Scenario	MBRL	MFRL	IL	Data-driven MPC	Justification (evidence)
Limited data; need planning look-ahead and sample efficiency	✓	✗	●	✓	Model learning with priors enables look-ahead with fewer samples than model-free RL; data-driven MPC likewise leverages logs for mismatch compensation while keeping constraints explicit [71], [78].
Broad variability / unstructured settings; high adaptability	●	✓	●	●	Model-free RL can internalize complex contacts/terrain variability when trained with broad randomization; primary risk is distribution shift and safety without explicit constraint layers (Table 21) [2], [74], [76].
High-quality demonstrations available; fast deployment needed	●	●	✓	●	IL reduces reward engineering and accelerates deployment when demonstrations cover the task; failure under unseen states motivates aggregation/safety scaffolding (Table 21) [58], [81].
Model errors dominate but logs are available for identification	●	●	●	✓	Error-model learning / adaptive MPC targets systematic residual dynamics while preserving constraint handling; typically easier to validate than end-to-end policies for safety-critical regimes [67], [79], [80].

are listed in Table 15, and usage guidance appears in Table 17.

2) LIMITATIONS

The main limitation of impedance-based approaches is their reliance on gain tuning and sensing quality. Overly stiff settings can undermine safety and make contact planners brittle; overly compliant settings can degrade tracking and destabilize locomotion. Moreover, impedance alone does not ensure feasibility: without an accompanying model-based layer (e.g., a QP enforcing friction cones), the controller may generate commands that are physically unrealizable.

Overall, impedance and force control are best viewed as necessary but not sufficient components: they complement WBC/MPC and learning-based policies by shaping interaction behavior, but must be integrated with explicit constraint enforcement and robust estimation to achieve reliable loco-manipulation.

Impedance and force-control layers most often fail due to gain mis-tuning or insufficient sensing/estimation quality, leading to either overly stiff unsafe interaction or overly compliant tracking loss (Table 20). Accordingly, the evidence motivates treating impedance/force control as a necessary interface for safe contact, but not sufficient on its own without explicit feasibility enforcement (QP/MPC) and robust state/contact estimation (Tables 20–21).

Taken together, Sections V-A-V-E provide a control-oriented view of loco-manipulation: each family offers different compromises between model fidelity, computational cost, safety guarantees, and adaptability. In the next section (Section VI) we step back from specific algorithms and instead organize the discussion around four recurring whole-body coordination challenges that all of these controllers must confront in practice.

F. EMERGING TRENDS (2023-2025): TOWARD UNIFIED, FORCE-AWARE LOCO-MANIPULATION

These trends are also reshaping the traditional separation between model-based, optimization-based, learning-based, and hybrid families. In particular, (i) *learning-in-the-loop* optimization (e.g., learned dynamics/residuals inside MPC/NMPC) blurs the optimization–learning boundary; (ii) *residual and safety-filtered learning* blurs learning–model-based boundaries by retaining explicit constraints via QPs/impedance; and (iii) *force-aware learning* narrows the historical gap between force-explicit WBC/MPC and force-implicit RL by treating interaction forces as first-class control targets. Table 6 provides a taxonomy-linked summary of why these boundary crossings are occurring: they are primarily responses to the dominant bottlenecks of each family (model mismatch, solver cost, safety, and distribution shift).

Very recent work has begun to blur the boundaries between the control families reviewed above and to scale loco-manipulation beyond single-task, hand-tuned setups.

First, multi-task whole-body policies trained with large-scale RL and curriculum design now tackle families of loco-manipulation tasks within a single controller, including combined locomotion, object interaction, and recovery behaviors. Examples include unified whole-body policies for quadruped mobile manipulators that jointly coordinate legs and arm in a single network [2], as well as cooperative loco-manipulation policies such as RoboDuet, where two legged agents learn a joint whole-body skill for collaborative transport and manipulation [81]. Data-driven loco-manipulation frameworks that operate on diverse human or teleoperated demonstrations for pushing, grasping, and carrying tasks [80] reinforce this shift from ‘‘one controller per scenario’’ toward reusable whole-body skill libraries.

TABLE 14. Representative hybrid loco-manipulation controllers (optimization ↔ learning).

Reference (Year)	Hybrid Strategy	Robot / Config.	Advantages	Limitations / Notes
Ma et al. (RAL'22) [10]	Learned locomotion policy + model-based manipulation (MPC/WBC); predicted wrench coupling	ANYmal + arm (sim/real)	Good sim-to-real; explicit disturbance handling between base and arm	Interface/tuning between modules; perception integration effort
Ferrolho et al. (2023) [5]	RL locomotion + MPC manipulation (RoLoMa)	Quadruped + arm	Robust whole-body coordination; handles arm-induced wrenches	Requires disturbance modeling; solver load for MPC
Kovalev et al. (2023) [60]	Predictive RL with MPC stabilization (gait and interaction)	Quadruped + arm	Stable planning with learned adaptability	Careful reward/cost co-design; training stability
Cheng et al. (2025) [61]	RL feedback + optimal control (RAMBO)	Quadruped + arm	Robust to unmodeled effects; balanced reactivity and constraints	Integration complexity; onboard compute

TABLE 15. Impedance/force-control approaches used within loco-manipulation stacks.

Reference (Year)	Approach	Robot / Config.	Reported Advantages	Notes / HRI relevance
Khatib (1987) [62]	Operational-space control with impedance targets	Generic manipulators	Direct task-space regulation; intuitive stiffness/damping	Foundation for task-level compliance in whole-body settings
Dietrich et al. (2016) [20]	Whole-body impedance control (survey/tutorial)	Human-centered robots	Passivity/compliance analysis; safe pHRI	Guidance for contact-consistent, redundancy-resolving compliance
Bellicoso et al. (2019) [52]	Posture/manipulability optimization with whole-body compliance (ALMA)	ANYmal + arm	Improved manipulability while keeping balance	Human-robot collaboration scenarios (handover, assistance)
Risiglione et al. (2022) [55]	Cartesian impedance via QP (task-space impedance in whole-body optimization)	HyQ + KUKA arm	Precise impedance shaping; constraint handling	Transparent interaction for quadruped manipulation
Portela et al. (2024) [59]	Learned whole-body force control	Quadruped + arm	Intuitive/compliant interaction; teleop-ready	Useful as a safety/post-shaping layer for learned policies

These results sit at the intersection of our learning-based and hybrid categories: training is predominantly RL- or demonstration-driven, but safety and torque limits still rely on model-based tracking layers.

Second, a series of works on learning force control for legged manipulation and on unified force/position policies for loco-manipulation show that contact forces are becoming first-class citizens in learning-based pipelines. Instead of treating force regulation as an emergent behavior, these approaches train policies that explicitly command or regulate contact forces while remaining compatible with impedance and WBC layers [59]. From the perspective of our taxonomy, this development directly addresses the "stability during dynamic manipulation" and "foot-ground contact modeling" challenges in Section VI, and it narrows the historical gap between model-based MPC/WBC (which handle forces explicitly) and RL (which traditionally did not).

Third, data-centric frameworks such as UMI-on-Legs-style systems combine real-world human demonstrations collected with hand-held grippers and robot-centric simulation data to train whole-body controllers for quadruped manipulator systems [80]. By decoupling task-dependent manipulation data collection from the robot platform itself, these approaches improve scalability and shift part of the effort from low-level control engineering to dataset design and offline training.

Taken together, these trends suggest that the field is moving toward (i) scalable, multi-task loco-manipulation policies; (ii) force- and contact-aware learning that directly addresses the coordination challenges in Table 22; and (iii) data pipelines that intertwine simulation, digital twins, and low-cost real-world demonstrations. In the conclusions (Section IX), we therefore emphasize research directions that explicitly exploit these emerging capabilities while retaining the safety and interpretability of model-based control.

Table 18 summarizes a set of very recent works (2023-2025) that extend or challenge the control families and task axes proposed in this review. By explicitly mapping each contribution to our taxonomy and to the coordination challenges in Section VI, the table makes clear how the field is shifting toward multi-task, force-aware, and data-centric loco-manipulation.

Overall, the dominant trajectory is toward *structured adaptivity*: retaining explicit feasibility/safety structure (contacts, torque/friction limits, impedance envelopes) while

TABLE 16. When to use what: **Hybrid controllers** (optimization ↔ learning).

Condition / Scenario	Planner+learned locomotion	Learned residual on model-based	Hierarchical (learned high-level + MPC/WBC)	RL feedback + optimal control	Justification (evidence)
Strong safety/feasibility requirements with known contact schedule	✓	●	✓	●	Allocate hard constraints to MPC/WBC while using learning for robustness or high-level decisions; mitigates safety limitations of pure RL noted in Table 20 [5], [10].
Large model uncertainty / variable payloads / rough contact	●	✓	●	✓	Residual learning and RL-augmented optimal control directly target mismatch and disturbance uncertainty; aligns with contact-uncertainty failure modes and mitigations (Tables 20–21) [60], [61].
Limited compute; need modular deployment across platforms	✓	✓	●	●	Modular split reduces online OCP burden while preserving feasibility; practical success reported for learned locomotion + model-based manipulation integration [10].
Frequent task switching and perception-driven goals (high-level decisions)	●	●	✓	✓	Learned high-level policies handle discrete decisions and perception-driven goal selection; MPC/WBC maintains low-level feasibility/safety, addressing latency/shift risks summarized in Table 21 [60], [61].

TABLE 17. When to use what: **Impedance / force-control** as cross-cutting layers.

Condition / Scenario	Task-space impedance in QP	End-effector admittance	Variable stiffness scheduling	Learned force residual	Justification (evidence)
Strict constraints (joint/torque/friction); multi-contact with arm+legs	✓	●	✓	●	QP-based impedance can enforce hard feasibility while shaping compliant interaction; consistent with operational-space and whole-body impedance foundations [20], [55], [62].
Limited torque sensing; external wrench sensing available (EE FT sensor)	●	✓	●	●	Admittance uses measured wrenches to generate motion references robustly when torque sensing is limited; widely used in pHRI compliance practice [20].
Human proximity tasks (handover, guidance); comfort prioritized	✓	✓	✓	●	Compliance and passivity-like behavior reduce impact/force peaks and improve comfort; reinforced by human-robot collaboration demonstrations on quadrupeds with arms [20], [52].
Teleoperation / rapid response to intent; residual adaptation to uncertainty	●	●	●	✓	Learned force residuals improve robustness to unmodeled contacts while impedance envelopes provide safety; aligns with force-aware learning trends [59].

adding data-driven components for perception, disturbance estimation, and residual compensation. This explains why recent contributions increasingly fall into the hybrid region of Table 6, even when their primary novelty is presented as "RL" or "MPC": the strongest empirical results often rely on explicit constraint layers for safety and on learned modules for robustness and generalization [5], [10], [59], [61].

G. TRADE-OFF MAPPING ACROSS CONTROL FAMILIES

To move from a method-by-method catalog toward a *critical synthesis*, representative works across model-based, optimization-based, learning-based, and hybrid stacks are compared using *quantitative anchors whenever explicitly reported in the cited papers*. The extracted anchors include: (i) evaluation setting (**Sim**, **HW**, **Sim+HW**), (ii) system scale (DoF; and horizon length for MPC when reported), (iii) real-time indicators (e.g., torque/policy/solver update rate in Hz, or stated real-time capability), and (iv) task-level outcomes (e.g., speed, success rate, payload/interaction forces, tracking/force errors) when explicitly stated. When a metric is not reported in a source, the entry is marked as NR (not reported) rather than inferred.

Table 19 summarizes *family-level* envelopes grounded in reported anchors from representative works, while Tables 20–21 synthesize *failure mechanisms* (why methods break) and *mitigations* supported by the literature. This evidence layer is then used to justify the selection guidance in the "when-to-use-what" tables throughout Section V.

Across families, the most pressing gaps are: (i) robust contact modeling and estimation under real terrain variability, (ii) predictable real-time performance for high-DoF whole-body optimization with rich perception in the loop, and

TABLE 18. Very recent loco-manipulation works (2023-2025) and how they reshape the control landscape defined in this review. Each work is positioned w.r.t. our control taxonomy (model-based / optimization-based / learning-based / hybrid), task axes (Section IV), and coordination challenges (Section VI).

Reference	Platform / Setting	Control family (this review)	Task axes primarily advanced	Coordination challenges addressed	Paradigm shift relative to classical WBC/MPC
Fu et al. (2023) [2]	Quadruped mobile manipulator (Go1 + arm, sim/real)	Learning-based (unified whole-body RL)	Interaction type (prehensile), Dynamics (dynamic), Sensing (proprio+vision)	Coupling between locomotion and manipulation; momentum/balance during arm motion	Single RL policy coordinating legs and arm, replacing decoupled locomotion+manipulation stacks; shows viability of unified whole-body policies under domain randomization.
Pan et al. (2024) [82]	Two legged agents (cooperative loco-manipulation)	Learning-based / hybrid (multi-agent RL + model-based tracking)	Coordination (multi-robot), Payload (medium-heavy), Dynamics (dynamic)	Coupling and stability during cooperative transport; momentum and balance in multi-agent settings	Extends loco-manipulation from single robot to cooperative whole-body behaviors, highlighting the need to rethink our coordination axis and control taxonomy for multi-agent scenarios.
Portela et al. (2024) [59]	Legged system performing manipulation with force feedback	Learning-based + impedance	Dynamics (dynamic), Terrain (uneven), Sensing (proprio+force)	Stability during dynamic manipulation; foot-ground contact modeling and control	Makes contact forces explicit decision variables in learned policies, narrowing the historical gap between force-explicit MPC/WBC and force-implicit RL.
Uehara et al. (2024) [81]	Quadruped with arm; large-scale human demonstrations + sim ("UMI on Legs")	Hybrid (IL/RL + model-based execution)	Interaction type (prehensile/non-prehensile), Sensing (proprio+vision), Coordination (single robot / human-robot)	Coupling between locomotion and manipulation; stability during human-in-the-loop demos	Shifts emphasis from hand-crafted controllers to data-centric pipelines (real demonstrations + simulation) for whole-body loco-manipulation, closely tied to benchmarking and reproducibility.
Recent NMPC + learning works (2023-2025) [71], [72]	Quadrupeds with manipulators (sim/early real)	Optimization-based + hybrid (NMPC with learned models/residuals)	Terrain (uneven/compliant), Payload (medium-heavy), Dynamics (dynamic)	Foot-ground contact modeling; momentum and balance control under model uncertainty	Integrate learned dynamics or residuals into NMPC for loco-manipulation, pointing toward compute-efficient hybrids that retain explicit constraints while adapting to unmodeled contacts and payloads.

(iii) standardized reporting of quantitative anchors (rates, horizons, success metrics, and failure cases) to enable reproducible benchmarking across platforms and tasks. These gaps motivate hybrid architectures that combine explicit feasibility enforcement with data-driven adaptation [10], [61].

VI. WHOLE-BODY COORDINATION CHALLENGES

Sections III-V introduced the modeling abstractions and control families (model-based, optimization-based, learning-based, and hybrids) used for quadruped loco-manipulation. In this section we change perspective: instead of organizing methods by mathematical tool or data usage, we organize the discussion around four recurring whole-body coordination challenges and use them as a lens to re-interpret the strengths and weaknesses of the control families. Specifically, we consider (i) coupling between locomotion and manipulation, (ii) stability during dynamic manipulation, (iii) foot-ground contact modeling and control, and (iv) momentum and balance regulation.

Developing robust and effective whole-body coordination in quadruped robots equipped with manipulator arms is still a big problem in loco-manipulation research. This section discusses the major technological difficulties confronting

TABLE 19. Evidence-based trade-off map across control families using *reported quantitative anchors* from representative works (NR = not reported in the cited paper).

Control family	Reported quantitative anchors (examples)	Evidence	Strengths supported by evidence	Typical breakdown modes (observed in practice)
Model-based ID-WBC / HQP	Torque-level whole-body control loop reported at **400 Hz** in ALMA [52] (others: NR)	HW	Strong constraint handling (torque/friction/contacts) and interpretable prioritization; reliable for structured interaction with calibrated models [20], [26], [52]	Sensitivity to model mismatch (mass/friction/contact location) and estimator latency; requires nontrivial tuning and margins [27], [65]
Whole-body MPC (unified locomotion+manipulation)	Average MPC update rate reported at **70 Hz** and horizon **T=1 s** for unified MPC on a quadruped+arm [1] (high-DoF WB-MPC: NR rate, **37 DoF** system scale reported [28])	Sim+HW / HW (varies)	Predictive contact and load-transfer reasoning; explicit trade-offs over a horizon; effective when contact timing and coupling matter [1], [28]	Solver overruns / infeasibility under fast contact switching and high uncertainty; computational burden grows rapidly with DoF and richer models [1], [28]
High-rate locomotion MPC (reduced-order / RF-MPC)	RF-MPC real-time operation reported at **250 Hz** for dynamic quadruped motions [29]	HW	High-rate stabilization of reduced-order locomotion dynamics; useful as a fast predictive layer beneath whole-body stacks [29]	Model reduction can underrepresent arm-induced wrenches and contact-rich manipulation; requires careful interfacing to whole-body torque control [29]
Hybrid MPC+WBC for highly dynamic motions	Top speed reported at **3.7 m/s** with MPC+whole-body impulse control on hardware [66] (other anchors: NR)	HW	Demonstrated robustness for highly dynamic locomotion by splitting planning (forces) and execution (whole-body feasibility) [66]	Complex tuning/handshakes across layers; failure can occur at mode transitions if contact assumptions diverge from reality [66]
Learning-based unified whole-body policies (RL/IL)	Policy execution at **50 Hz** reported for unified whole-body RL on a quadruped+arm (sim-to-real) [2]	Sim+HW	High adaptability to unstructured variation when training coverage is broad; reduces dependence on exact analytic models [2], [58]	Distribution shift and safety/constraint violations without explicit constraint layers; performance depends on data coverage and sim-to-real fidelity [2], [58], [81]
Hybrid learning+model-based stacks	(Reported anchors vary by design; often NR for rates in survey-level reporting) Examples include RL+MPC/WBC integration for disturbance-aware coordination [5], [10], [61]	Sim+HW / HW (varies)	Best empirical balance of constraint enforcement (model-based) with adaptation (learned residuals/policies); mitigates dominant failure modes of pure families [10], [61]	Increased software/verification complexity; end-to-end guarantees remain difficult when learned modules change [60], [61]

such systems, including coupling between locomotion and manipulation, stability during dynamic manipulation, foot-ground contact modelling, and momentum and balance control, summarized in Table 22. These obstacles must be handled holistically in order to do complicated actions such as carrying, pushing, or engaging with things while walking across uneven terrain.

In what follows, each subsection first summarizes the challenge and then explicitly comments on how the different control families of Section V fare with respect to it. This turns the challenges from a stand-alone catalog into an evaluative lens for the survey.

A. COUPLING BETWEEN LOCOMOTION AND MANIPULATION

The control of quadruped robots with arms involves tightly coupled kinematics and dynamics between the base (responsible for locomotion) and the manipulator (responsible for task-space interaction). Traditional modular architectures often treat these systems separately, resulting in suboptimal behavior. Li et al. [64] proposed a unified inverse dynamics-based whole-body controller for a torque-controlled quadruped robot with an arm. This method explicitly considers the interaction between the base and the manipulator in a coupled optimization problem. The resulting formulation enables coordinated motion planning and task execution across the full robot structure, improving overall task efficiency and compliance. Additionally, the work by Dadiotis et al. [28] further explores this coupling in a real-time MPC framework applied to a dual-arm quadruped with 37 DoF, emphasizing real-time feasibility (cf. (30)-(31)).

Viewed through the taxonomy in Section V, coupling is handled most directly by whole-body RBD models and controllers: ID-WBC (Section V-A) enforces coupling at the acceleration/force level, while WB-MPC (Section V-B) predicts its effect over a horizon. Both can, in principle, achieve tight base-arm coordination for prehensile and non-prehensile tasks, but both inherit the modeling and computational demands of the full dynamic model (Section III). Modular or decoupled designs-common in learning-based and hybrid stacks (Sections V-C and V-D)-alleviate these demands by letting a locomotion policy treat arm forces

TABLE 20. Failure-mode synthesis (Part I): Task physics & interaction conditions.

Task condition	Failure manifestation (what breaks)	Most affected families	Mitigations supported by cited works
High contact uncertainty (friction variation, compliant terrain)	QP/OCP infeasibility; slip; base drift; degraded force tracking; conservative saturation reduces performance	ID-WBC, WB-MPC	Friction margins and contact blending; disturbance-aware prediction; task-space impedance shaping; learned residuals for unmodeled effects [27], [56], [61], [65]
Fast contact switching / dynamic manipulation while walking	Tracking loss at mode transitions; solver overruns; instability during rapid gait changes	Schedule-based WBC, WB-MPC	Hierarchical decomposition (long-horizon planning + short-horizon stabilization); reduced-order predictive layers; hybrid stacks offloading fast stabilization [29], [57], [66]
Large arm-induced wrenches / uncertain payload during locomotion	Base attitude drift; contact constraint violation; foot slippage under load transfer	WBC, MPC without compensation	Disturbance-compensation MPC; explicit wrench modeling/estimation; passive/compliant manipulation to reduce disturbance injection [5], [56], [73]
Strict safety requirements (HRI, explicit torque/friction limits)	Unsafe forces/impacts; saturation-induced oscillations; certification gaps for closed-box policies	Pure RL / IL	Explicit constraint layers (QP/MPC) around interaction; impedance/force shaping; hybrid "safety envelopes" (planning + compliant execution) [10], [20], [26], [59]

TABLE 21. Failure-mode synthesis (Part II): System-level constraints & deployment realities.

Task condition	Failure manifestation (what breaks)	Most affected families	Mitigations supported by cited works
Perception/state-estimation latency (vision-conditioned goals, delayed contact events)	Delayed recovery; oscillations; "late" constraint enforcement; brittle contact replanning	MPC and perception-heavy RL/IL	Robust estimation pipelines; conservative constraints; hierarchical planning; safe fallback layers (e.g., impedance + saturation + feasibility filters) [1], [10], [60]
Compute-constrained onboard deployment (embedded CPU/GPU limits)	Reduced horizon/solver accuracy; control-rate drops; missed real-time deadlines	WB-MPC, heavy hybrids	Model reduction; warm-starting and tailored solvers; hierarchical MPC; separating fast torque feasibility (WBC) from slower prediction (MPC) [1], [29], [66]
Distribution shift (new terrains, objects, lighting; unseen contacts)	Policy collapse; brittle perception coupling; degraded success under unseen settings	Vision-conditioned RL/IL	Domain randomization; data-centric pipelines; structured low-level tracking; hybrid residual adaptation [2], [58], [81]
Limited data / demonstration coverage	Overfitting; poor generalization; failure in rare contact states	IL, data-driven MPC	Demonstration aggregation and curriculum design; model-based priors for sample efficiency; combining real demonstrations with simulation scaling [71], [78], [81]

as disturbances and the arm treat the base as a moving support. These are often easier to implement but struggle when the coupling itself is task-critical (e.g., co-carrying heavy payloads or using body motion to extend arm reach).

In summary, strong locomotion-manipulation coupling tends to favor model-based and optimization-based families (WBC, WB-MPC), while learning-heavy or decoupled hybrids are preferred when approximate coupling is sufficient and robustness to unmodeled effects is more important than exact whole-body coordination.

B. STABILITY DURING DYNAMIC MANIPULATION

Dynamic manipulation, such as lifting or interacting with moving objects, can generate destabilizing forces on the robot's base. Ensuring whole-body stability requires predicting these disturbances and compensating for them during motion. Rigo et al. [82] introduced a hierarchical optimization framework that integrates manipulation planning into the whole-body control loop. Using a dual-layer MPC structure, they optimize both contact force regulation and motion generation, enabling the robot to manage

dynamic interaction forces without compromising stability. Risiglione et al. [55] also contribute to this domain with a simplified impedance-based whole-body controller that facilitates compliant responses to manipulation-induced perturbations. Fahmi et al. [83] validated a passive control strategy that maintains stability over rough terrain.

Dynamic stability exposes a different set of trade-offs across control families. Model-based WBC can regulate contact forces and trunk motion effectively in quasi-static regimes but may become reactive and myopic when large arm motions and fast contact changes are involved. Optimization-based MPC, especially disturbance-compensation and hierarchical variants (Section V-B), explicitly anticipates how manipulation wrenches will affect future base states and thus scores well on this challenge when sufficient compute and reasonably accurate models are available. Learning-based controllers (Section V-C) can, in principle, internalize complex interaction patterns and achieve impressive robustness, but their behavior under out-of-distribution perturbations is harder to certify and debug. Hybrids that combine MPC or WBC with learned residuals (Section V-D) attempt to get the best of both worlds: analytic stability margins with data-driven adaptation to unmodeled disturbances.

Thus, for highly dynamic loco-manipulation tasks, optimization-based and hybrid approaches tend to be the most compelling, while pure WBC is better suited to slower or more structured interactions, and pure learning remains attractive when data is abundant but formal stability guarantees are less critical.

C. FOOT-GROUND CONTACT MODELING AND CONTROL

Ground contact forces significantly impact a quadruped's ability to maintain stability, especially during arm movement. Shi et al. [84] proposed a multi-objective torque control framework that addresses both motion tracking and contact force regulation in hydraulic quadrupeds. Their method compensates for model inaccuracies by integrating external force estimation and adaptive control laws, improving robustness across different terrains. Similarly, Wolfslag et al. [85] optimized body-ground contact points to augment whole-body manipulation capabilities, while Lu et al. [86] focused on terrain-adaptive motion planning and contact reaction force generation.

Contact modeling sits exactly at the intersection of the dynamics in Section III and the control methods in Section V. Model-based WBC and MPC explicitly embed friction cones and unilateral constraints (Eqs. 3, 29) and are therefore the natural tools when contact feasibility and terrain variability (Section IV) are central. They are particularly strong on uneven or compliant terrain but are sensitive to errors in contact position and friction estimates. Learning-based policies, especially those trained with heavy domain randomization, can acquire surprisingly robust contact strategies without explicit modeling, but they typically require a separate safety filter or impedance layer (Section V-E) to avoid generating unsafe contact forces on hardware. Hybrid controllers that learn residuals on top of analytic contact models offer a principled compromise: the model guarantees basic feasibility, while learning accounts for unmodeled terrain effects and hardware idiosyncrasies.

In terms of performance vs. guarantees, foot-ground contact challenges thus push the field toward either (i) model-centric WBC/MPC with carefully calibrated contact models or (ii) hybrid designs where learning augments but does not replace contact consistency constraints.

D. MOMENTUM AND BALANCE CONTROL

Maintaining whole-body balance in response to dynamic motions and external disturbances is essential for functional loco-manipulation. Morlando et al. [87] proposed a control architecture that incorporates a momentum-based observer into the whole-body controller. This allows for real-time disturbance rejection through feedback on centroidal momentum. Lee et al. [88] extended this concept by incorporating torque sensors for angular momentum regulation. Bellicoso et al. [89] developed a unified locomotion and whole-body control strategy for quadrupeds that stabilizes both the robot's trunk and legs across dynamic gaits.

Momentum and balance control are closely tied to the centroidal dynamics of Section III and are a key benchmark for comparing controllers. Centroidal/MPC schemes (Section V-B) are explicitly designed around $\dot{h}_G$ and CoM trajectories and therefore handle this challenge naturally, especially in dynamic gaits and when non-prehensile interactions or terrain variations are present. ID-WBC can also regulate momentum via task hierarchies but tends to emphasize posture and contact force consistency more than long-horizon balance margins. Learning-based methods excel when disturbances and terrain variations are too complex to model but require sufficient training coverage to avoid catastrophic balance failures in rare cases. Hybrid approaches-for example, momentum-based observers with RL policies or MPC-can improve robustness by feeding back estimated disturbances into both model-based and learned components.

Overall, momentum and balance control highlight one of the main themes of this review: reduced-order, centroidal models and MPC provide a strong backbone for dynamic balance, while whole-body RBD models and impedance layers are essential to ground these abstractions in physically realizable torques and contact forces, and learning/hybrid methods offer adaptability beyond what analytic models can capture.

Putting these four challenges together, Table 22 can be read not only as a list of problems and techniques but also as a qualitative scorecard for the control families of Section V. Model-based ID-WBC scores highly on tight coupling and contact consistency but is limited in dynamic, rapidly changing scenarios. Optimization-based MPC is strongest on dynamic stability and momentum regulation but is compute- and model-hungry. Learning-based controllers offer adaptability for all four challenges but need data

and safety scaffolding. Hybrid methods attempt to occupy the middle ground by assigning different aspects of each challenge to the components best equipped to handle them.

Many of these challenges are difficult and risky to study exclusively on hardware, especially when exploring failure modes or aggressive maneuvers. As a result, high-fidelity simulation and digital-twin infrastructure play a central role in the development and validation of loco-manipulation controllers; Section VII reviews the main platforms and their trade-offs.

Table 22 turns the coordination challenges into an evaluative lens: each challenge is tied to (i) measurable indicators used in prior work, (ii) dominant failure triggers, and (iii) evidence-backed trade-offs across control families. This mapping is used throughout the remainder of Section VI to justify why different method families succeed or fail under specific task conditions (terrain uncertainty, interaction dynamics, payload shifts, and latency), and to motivate the shift toward hybrid architectures that combine explicit feasibility with data-driven adaptation.

VII. SIMULATION ENVIRONMENTS AND DIGITAL TWINS

Building on the coordination challenges discussed in Section VI, this section focuses on the simulation and digital-twin tools that enable safe exploration, tuning, and benchmarking of loco-manipulation controllers before deployment on physical quadruped platforms.

Because simulator fidelity and digital-twin calibration depend on accurate robot parameters (mass, payload, arm inertia, DoF, and actuation limits), Table 23 summarizes representative quadruped–arm platforms used in loco-manipulation studies to contextualize what must be modeled for transferable results.

A. HIGH-FIDELITY SIMULATION PLATFORMS

In loco-manipulation, simulation is not only a prototyping tool but also a mechanism for stress-testing contact-rich failure modes and for calibrating model assumptions via digital-twin replay.

1) ISAAC SIM

Isaac Sim (Omniverse/PhysX) is widely used for contact-rich legged simulation, synthetic vision data, and large-scale domain randomization pipelines via Isaac Gym/Lab [95], [96]. Its practical advantage for loco-manipulation is the ability to co-simulate *two contact channels* (feet and end-effector) alongside realistic sensing/latency, which is particularly valuable for hybrid pipelines.

2) GAZEBO

Gazebo remains a standard choice for ROS-centric integration, HIL workflows, and regression testing of full stacks [97]. For loco-manipulation, its main limitation is that contact/force realism depends strongly on the selected physics engine and plugin design, so sim-to-real success often hinges on careful calibration.

3) MUJOCO

MuJoCo provides fast, stable rigid-body contact simulation and is therefore common for RL and trajectory optimization benchmarking [98], [99]. However, digital-twin synchronization and high-fidelity perception pipelines typically require additional infrastructure beyond the core simulator.

4) SIMULATION FIDELITY REQUIREMENTS ACROSS CONTROL FAMILIES

Simulation fidelity requirements differ systematically across controller families in Section V. **Model-based WBC** and **optimization-based MPC** require accurate rigid-body parameters, contact geometry, friction, and estimator delay models because constraint satisfaction and feasibility margins are computed from these quantities; small mismatches can yield QP/OCP infeasibility or force-tracking errors under contact [27], [28]. In contrast, **learning-based** policies are less sensitive to a single "best" model but require *broad variability coverage* (domain randomization over masses, friction, compliance, delays, and sensing) to prevent collapse under distribution shift [2], [22]. **Hybrid** stacks combine both requirements: the model-based core demands calibrated feasibility, while the learned component demands variability coverage and safe interfaces; digital-twin replay and parameter tuning are therefore particularly valuable for debugging hybrid handshakes [10], [61], [100].

From a loco-manipulation standpoint, the most fidelity-critical elements are (i) *two-contact-channel realism* (feet–ground and end-effector–object), (ii) *arm-induced wrench coupling* into base stability, and (iii) *latency-aware sensing*, since perception and state estimation delays directly affect both force safety and contact feasibility during manipulation while walking [56], [59].

B. DIGITAL TWINS FOR REAL-TIME FEEDBACK AND CONTROL TUNING

A digital twin is a real-time synchronized virtual model of a physical robot that mirrors its kinematic states, sensor feedback, and actuator commands. Digital twins provide the ability to conduct predictive simulations and preemptively identify failure conditions. In quadruped robots with arms, this capability is instrumental in tuning impedance parameters, refining MPC policies, and validating behavior under simulated perturbations.

For example, Ali et al. [100] developed a digital twin framework that integrates sensor fusion data from a real robot with a high-fidelity simulated counterpart. This architecture enabled the robot to autonomously adapt to changes in surface friction and payload mass by leveraging simulation-informed adjustments. Similarly, Naskar et al. [101] demonstrated digital twin-based planning and obstacle avoidance, allowing for efficient trajectory replanning when encountering unexpected terrain discontinuities.

TABLE 22. Evidence-based coordination scorecard for quadruped loco-manipulation. For each challenge, the table lists (i) measurable indicators used in the literature, (ii) dominant failure triggers, (iii) control-family trade-offs, and (iv) quantitative anchors (only when explicitly reported in the cited works).

Coordination challenge	Measurable indicators (examples)	Dominant failure triggers	Control-family trade-offs (critical synthesis)	Quantitative anchors (reported)
Coupling between locomotion and manipulation	EE pose/force error; base drift during manipulation; CoM tracking error; torque/GRF margin under arm motion	Strong base–arm coupling (large wrenches, payload shifts); simultaneous contact transitions + arm motion; calibration/estimation errors	**ID-WBC:** tight coupling at force/acceleration level and strong constraint handling, but sensitive to model/estimation mismatch and tuning burden. **WB-MPC:** anticipates coupling over a horizon, but compute- and model-hungry. **RL/IL:** can internalize coupling if trained broadly, but weaker hard-constraint guarantees. **Hybrids:** most reliable when analytic feasibility (WBC/MPC) is paired with learned disturbance adaptation.	CENTAURO whole-body MPC demonstrated on a **37-DoF** platform [28]; unified whole-body RL executed at **50 Hz** in a legged manipulator stack [2].
Stability during dynamic manipulation	Peak trunk tilt/ang. velocity; recovery time after interaction; contact-force overshoot; task success under disturbances	Fast arm-induced impulses while walking; delayed perception/estimation; mode-switch transients (lift/push/carry); solver overruns (missed deadlines)	**WBC:** strong for quasi-static/structured interactions; becomes reactive when disturbances are fast and contact switches are frequent. **MPC variants:** disturbance-compensation and hierarchical MPC improve stability by anticipating interaction wrenches, but require compute and robust estimation. **Learning:** can be highly robust to disturbances, but certification/debugging under rare perturbations remains difficult. **Hybrids:** use MPC/WBC as a safety envelope while learning captures unmodeled interaction dynamics.	Centroidal impulse-control baseline illustrates typical real-time splits (e.g., **WBC** $\sim$**0.5 kHz** with **MPC** $\sim$**40 Hz**) [66]; disturbance-compensation MPC explicitly models arm-induced wrenches [56]; hierarchical optimization for interaction-aware stability [83].
Foot–ground contact modeling and control	Slip rate / foothold loss; GRF tracking error; constraint violations (friction cone, unilateralarity); infeasible QP/OCP occurrences	Unknown/variable friction; terrain compliance/discontinuities; contact-location uncertainty; sensing/estimation bias	**WBC/MPC:** explicit friction/unilateral constraints offer principled feasibility, but performance degrades with friction/terrain mismatch. **Learning:** robust contact strategies possible via domain randomization, but typically needs safety filters/impedance shaping on hardware. **Hybrids:** keep explicit contact feasibility while learning residuals for unmodeled terrain effects (often the most practical compromise).	High-rate contact-aware MPC is feasible in reduced-order formulations (e.g., **250 Hz** representation-free MPC for locomotion) [29]; passive/robust whole-body control validated on challenging terrain [84]; multi-objective torque/contact regulation in hydraulics [85].
Momentum and balance regulation	Centroidal momentum error; CoM tracking error; trunk stabilization (roll/pitch/yaw); GRF distribution consistency	External pushes + interaction forces; payload-induced inertia shifts; contact timing errors; estimator noise	**Centroidal MPC:** naturally aligned with momentum/CoM objectives and anticipatory balance margins, but depends on model quality and compute. **ID-WBC:** effective via momentum tasks/priors, but tends to be more local (myopic) than horizon-based planning. **Learning:** can handle complex disturbances, but needs broad training coverage and safety scaffolding. **Hybrids:** combine momentum observers + predictive planning with learned adaptation for unmodeled disturbances.	Impulse-control + centroidal MPC as a quantitative reference point (**MPC** $\sim$**40 Hz**, **WBC** $\sim$**0.5 kHz**) [66]; momentum-observer and trunk-stabilization pipelines [88], [89]; dynamic locomotion stabilization foundations [90].

C. BENCHMARKING AND REPRODUCIBILITY CONCERNS

Benchmarking provides a consistent framework to evaluate the generalization, agility, and robustness of loco-manipulation controllers. Standardized scenarios such as Barkour [102] evaluate a quadruped's ability to execute animal-like maneuvers including jumps, sprints, and sharp turns, which are especially difficult when coupled with manipulation tasks. Stanford Pupper [103] offers a more accessible benchmarking platform aimed at educational and open-source reproducibility goals.

In addition to locomotion-only benchmarks, recent high-speed legged-manipulation demonstrations underscore that future benchmark suites must include *contact-rich, coupled* leg–arm regimes and report force- and feasibility-related metrics, not only task success [104].

Despite progress, reproducibility in simulation-based robotics research is still a significant concern. Spencer et al. [105] outlined the pitfalls of variation in simulation configurations, hardware-software synchronization issues, and lack of open-source baselines. They emphasized the importance of containerized environments, digital twin validation, and remote competition-based evaluations to standardize research.

High-fidelity simulation and digital twins are indispensable for modern quadruped manipulation systems. Their role extends beyond prototyping to real-time feedback, behavior validation, and robust benchmarking. As robotic systems become more complex, the co-evolution of simulation environments and reproducibility standards will be crucial for the field's advancement.

Table 24 summarizes platform-level trade-offs, but the key point for loco-manipulation is that *contact and force realism* (feet and end-effector) and *latency-aware sensing* often dominate transfer performance more than nominal "physics accuracy" labels. Accordingly, the remainder of this section shifts from listing simulators to analyzing (i) which fidelity dimensions each control family requires and (ii) which task regimes remain hardest to transfer.

Linking simulation fidelity to taxonomy and control families. The regimes that are hardest to transfer in

TABLE 23. Representative quadruped–arm platforms reported in loco-manipulation studies (parameters relevant for simulation/digital-twin modeling and sim-to-real calibration).

Four-Legged Robot Name	Robot's Weight (kg)	Payload Capacity (kg)	Manipulator Arm	DoF	Arm's Weight (kg)	Steady-State Load Capacity (kg)	Planning Approach	Operational Scenarios
A1 [15]	12	5	WidowX 250	6	2.63	0.25	SS	SaR
Vision 60 [91]	51	10	Kinova Mico 2	4	3.8	5.2	CS	-
ANYmal [92]	30	15	Kinova Jaco 2	6	4.4	1.6	SS/CS	Object carrying and pulling
ANYmal C [1]	30	15	DynaArm (Custom-made)	4	-	7	CS	Object transport/maneuver
Aliengo [93]	21.5	13	Unitree Z1	6	4.3	2	SS	Fire rescue
Aliengo [94]	21.5	13	ViperX 300	5	3.63	0.75	CS	Opening/doors
Spot [5]	32.7	14	Spot Arm	5	8	5	CS	Object transport/maneuver
Self-designed [3]	25	-	Kinova Jaco 2	6	4.4	1.6	CS	-
Self-designed [12]	205	-	Self-designed	5	35	24	SS	Grasping
Spot [95]	32.7	14	Kinova Gen3	7	8	4	SS	Dynamic grasping

loco-manipulation are those that simultaneously stress (i) the *foot–ground contact channel* (uncertain friction/compliance and contact timing), (ii) the *end-effector contact channel* (impact/force regulation and object properties), and (iii) *latency/partial observability* in state estimation and perception. In these regimes, **model-based WBC** and **optimization-based MPC** are typically the most sensitive to sim-to-real gaps because they compute contact forces and enforce feasibility through explicit friction cones, unilateral constraints, and predictive dynamics; small errors in friction/contact geometry or delay can induce force-tracking errors, slip, or QP/OCP infeasibility [27], [28], [56]. **Learning-based** approaches are less sensitive to any single nominal model but are highly sensitive to *distribution shift* if training does not cover contact/latency variability (e.g., micro-slip, object compliance, actuator saturation, perception delay), motivating domain randomization and real-data adaptation [2], [22]. **Hybrid** stacks can improve transfer by combining hard feasibility/safety guarantees from WBC/MPC/impedance with learned residuals or high-level decisions that compensate mismatch [10], [61]. This motivates the taxonomy-aligned transfer analysis in Table 27 (Table 24).

Table 27 highlights that the hardest regimes to transfer are not defined by locomotion or manipulation alone, but by their *simultaneous contact coupling*: feet–ground contact uncertainty compounds end-effector contact uncertainty, while arm-induced wrenches perturb the base precisely when footholds are changing. This coupling amplifies small modeling and latency errors into large whole-body tracking deviations, explaining why dynamic loco-manipulation on uneven terrain and force-limited HRI remain the most transfer-challenging settings. As a result, increasing simulator fidelity alone is insufficient; robust transfer typically requires a combination of calibrated model structure (for feasibility and safety) and variability coverage or residual adaptation (for unmodeled effects), which is consistent with the growing dominance of hybrid approaches in recent loco-manipulation systems [10], [61].

Having summarized platform capabilities and family-specific fidelity needs, we now focus on the remaining bottleneck: how mismatch in contact, actuation, sensing, and perception manifests during transfer for different controller families and task regimes.

D. SIM-TO-REAL GAP AND CONTROL-FAMILY TRADE-OFFS

Even with high-fidelity physics engines and carefully constructed digital twins, there is always a discrepancy between simulated dynamics and the real robot. This *sim-to-real gap* is especially critical for learning-based and data-driven methods that rely on large-scale simulation for training. However, it also affects model-based WBC and MPC, which are sensitive to inertial parameters, contact properties, and sensor latency. In this subsection we discuss, for each control family in Section V, (i) how simulation/digital twins are typically used, (ii) common failure modes when transferring to hardware, and (iii) mitigation strategies and trade-offs. Table 26 summarizes the discussion.

1) MODEL-BASED (WBC / OPERATIONAL-SPACE CONTROL)

For classical and inverse-dynamics WBC, simulation is primarily used for controller design, gain/priority tuning, and validation under different terrains and payloads before deployment on hardware. Because these controllers explicitly depend on the rigid-body and contact models (e.g., $M(q)$, $C(q, \dot{q})$, $G(q)$, $J_c(q)$), the dominant sim-to-real failure modes

TABLE 24. Comparative assessment of simulation and digital-twin platforms for quadruped loco-manipulation. The added column summarizes each platform's *contact/force fidelity and sim-to-real utility* for contact-rich loco-manipulation (feet and end-effector).

Platform	Physics accuracy	Manipulator support	Digital twin ready	RL integration	Benchmark use	Contact/force fidelity + sim-to-real notes (loco-manip.)
Isaac Sim [96], [97]	High	Yes	Yes	Strong (Isaac Gym/Lab)	Growing	Strong for contact-rich training with domain randomization and sensor realism; suitable for vision + contact pipelines; benefits hybrids that mix learning with constraint layers.
Gazebo [86], [98]	Moderate	Yes	Partial	Moderate	Widely used	Good ROS/HIL integration; contact fidelity depends on engine/plugins; best for software integration/regression tests more than micro-contact transfer.
MuJoCo [99], [100]	Very high	Limited (modeling effort)	No	Excellent	Research-grade RL	High-quality rigid contact dynamics and fast rollouts; strong for RL/TO research but vision/digital-twin workflows require extra infrastructure.
Webots [107]	Moderate	Yes	No	Basic	Educational	Useful for prototyping and teaching; limited contact/force realism for contact-rich loco-manipulation transfer.
Unity Robotics [108]	Low (physics)	Visual only	No	Weak	Visualization only	Best for photorealistic perception datasets; not suitable as a standalone dynamics/contact simulator for force-critical tasks.
Custom digital twins [101], [102]	Platform-dependent	Yes	Yes	Yes	Case-specific	Best for sim-to-real calibration via log replay and parameter identification; high value for model-based/MPC stacks sensitive to inertial/contact mismatch.

are: (i) inaccurate inertial parameters, (ii) mis-modeled contact locations and friction coefficients, (iii) unmodeled actuator dynamics and joint backlash, and (iv) state-estimation delay and noise.

Mitigation strategies are largely *model-centric*: system-identification of inertial parameters, offline and online calibration of contact/friction models, adding safety margins to friction cones and torque limits in the QP, and designing passivity-/robustness-inspired WBC layers that tolerate bounded model errors. [65], [89] Digital twins are especially useful here: they provide a structured way to reconcile simulation and hardware by replaying real logs in the simulator and iteratively adjusting model parameters until predicted and measured contact forces and base motion match within acceptable bounds. [100], [101]

2) OPTIMIZATION-BASED (MPC / TRAJECTORY OPTIMIZATION)

For MPC and trajectory optimization, simulation is used not just for tuning but also for stress-testing long-horizon behaviors, validating contact schedules, and benchmarking solver performance. [28], [29] Here, the sim-to-real gap manifests in three main ways: (i) model mismatch accumulated along the prediction horizon (leading to poor contact timing or foot placement), (ii) inaccurate terrain models and friction parameters (causing unexpected slipping or loss of contact feasibility), and (iii) estimator/sensor delays that are not accounted for in the prediction model.

Mitigation strategies extend those used in WBC: robust or tube-based MPC formulations that inflate constraints to account for model uncertainty, disturbance observers and momentum-based estimators that feed back unmodeled wrenches into the MPC, [56] and hierarchical MPC structures where conservative high-level plans are tracked by lower-level controllers that compensate for fast disturbances. [57], [69] Digital twins support "closed-loop" testing where real sensor streams drive the simulated model, helping identify where prediction errors accumulate and guiding model refinement.

3) LEARNING-BASED (RL, IL, AND MBRL)

For reinforcement learning and imitation learning, simulation is usually the *primary* training environment: learning directly on hardware is often too risky and sample-inefficient. [22] The main failure modes when transferring policies trained in simulation to a physical quadruped with an arm include:

TABLE 25. Hardest-to-transfer loco-manipulation regimes (Table 24), explicitly mapped to the taxonomy axes in Section IV and to control-family sensitivity in Section V. The key transfer bottleneck is *simultaneous uncertainty* in both contact channels (feet and end-effector) under latency/partial observability.

Task regime (taxonomy-aligned)	Dominant sim gap mechanism	Most sensitive control families (why)	Typical hardware failure manifestation	Mitigations reported / suggested
Dynamic manipulation while walking (dynamic; contact-rich; coupled leg–arm)	Arm-induced wrench coupling + fast contact switching; prediction drift from friction/contact mismatch [28], [56]	**WBC/MPC: High** (constraints/feasibility depend on friction, contact timing, delays); **Learning: High** if contact/latency not covered (shift); **Hybrid: Medium–High** (inherits both, but can bound risk)	Base drift, slip, missed grasps, solver overruns or unsafe transients during mode switches	Disturbance-aware MPC; hierarchical MPC/WBC; conservative feasibility margins; task-space impedance shaping [55]–[57]
Uneven/compliant terrain with interaction (terrain uncertainty; contact-rich)	Foot micro-contact and friction/compliance mismatch dominate; GRF distribution differs from sim [27], [65]	**WBC/MPC: High** (foot contact model errors violate cones/unilaterality); **Learning: Medium–High** (needs friction/compliance randomization); **Hybrid: Medium** (robust core + residuals)	Unexpected slipping, contact loss, degraded end-effector force tracking	Contact blending/margins; passivity/robust WBC; domain randomization over friction/compliance; residual adaptation [2], [61], [65]
HRI / handover and force-limited interaction (strict safety; compliance)	Force/impedance realism + latency-aware sensing; safety constraints depend on timing and contact estimation [20], [59]	**WBC/impedance: High** (force envelopes must be correct); **MPC: Medium–High** (delay harms predictive safety); **Learning: High** without safety wrappers	Force overshoot, discomfort/unsafe impacts, saturation-induced instability	Explicit impedance envelopes; hard constraints; safety filters around learning; latency/noise modeling [10], [20], [59]
Vision-conditioned object interaction (vision; partial observability)	Photometric/geometric domain gap + perception latency/occlusion [2], [58]	**Learning: High** (perception shift); **WBC/MPC: Medium** (if relying on perception for contacts/goals); **Hybrid: Medium** (fallback controllers reduce risk)	Approach/grasp failures, delayed reactions, policy collapse under occlusions	Synthetic data + domain randomization; real demos; latency/noise injection; hybrid fallback [2], [81]
Quasi-static pick/place on flat ground (structured; low dynamics)	Lower coupling; more repeatable contacts; residual inertial/friction mismatch	**WBC/MPC: Medium** (calibration helps); **Learning: Medium** (less shift if task is narrow)	Moderate tracking error, minor force bias	Calibration and friction-cone margins; log-replay digital-twin tuning [101], [102]

(i) dynamic mismatch (mass, inertia, actuator limits, joint friction), (ii) contact and terrain mismatch (true friction, compliance, micro-geometry), (iii) sensor and actuator latency not modeled in simulation, and (iv) perception gap (lighting, textures, occlusions) for vision-based policies.

Consequently, sim-to-real for learning-based loco-manipulation is dominated by three mitigation patterns:

- *Domain randomization*: randomizing masses, friction, gains, delays, and terrain during training so that the learned policy is robust to a range of plausible hardware conditions. [2]
- *Real-to-sim calibration and residual learning*: first identifying a dynamics model or residual from real data, then training the policy in a calibrated simulator or on top of a residual dynamics term. [71], [79]
- *Real-world fine-tuning and safety wrappers*: starting from a sim-trained policy, then performing limited on-hardware adaptation with strict safety constraints (e.g., torque limits, impedance envelopes, and fallback WBC/MPC controllers). [59], [80]

Recent large-scale systems such as UMI-on-Legs-style frameworks use diverse real-world demonstrations combined with simulation to reduce the gap between training and deployment, emphasizing data diversity and task coverage rather than simulation fidelity alone. [80]

4) HYBRID CONTROLLERS (LEARNING + MODEL-BASED)

Hybrid controllers exploit simulation in two complementary ways: (i) to tune and validate the model-based components (WBC/MPC) using conventional workflows, and (ii) to train the learned components (policies, residuals, or cost-shaping networks) with domain randomization and calibrated models. [5], [10], [60], [61] Their sim-to-real failure modes are therefore a combination of the above: if the learned

TABLE 26. Sim-to-real gap patterns and mitigation strategies across control families.

Control family	Primary role of simulation / digital twin	Dominant sim-to-real failure modes	Common mitigation strategies / representative works
Model-based (WBC, operational-space control)	Offline design, gain/priority tuning, contact-model validation; digital twin used for parameter calibration and regression testing.	Inertial and friction parameter errors; mis-modeled contact locations; unmodeled actuator dynamics; estimator latency and noise.	System identification and friction calibration; robust/passive WBC design; safety margins in QP constraints; digital-twin-based replay and parameter tuning. [65], [90], [101], [102]
Optimization-based (MPC, trajectory optimization)	Validation of long-horizon behaviors, contact schedules, and solver performance; hardware-in-the-loop tests with digital twins.	Prediction drift due to model mismatch; incorrect friction/terrain models; unmodeled delays in state estimation and actuation.	Robust/tube MPC; disturbance observers and momentum-based estimation; hierarchical MPC (coarse conservative plans + reactive low-level control); digital-twin testing under recorded disturbances. [28], [29], [56], [57], [69]
Learning-based (RL, IL, MBRL)	Primary training environment (large-scale rollouts); exploration of rare events; data generation for vision and force policies.	Dynamics and contact mismatch; sensor and actuator latency; perception gap (lighting, textures, occlusions); distribution shift beyond training randomizations.	Heavy domain randomization and noise injection; calibrated dynamics models and residual learning; limited real-world fine-tuning with safety wrappers (WBC/MPC or impedance); use of diverse real-world demonstrations. [2], [22], [59], [71], [81]
Hybrid (learning + model-based)	Joint use: simulation for model-based tuning and for training learned components; digital twins for validating interaction between learned and analytic modules.	Learned module pushes model-based core into poorly modeled regimes; conservative model-based layer cancels beneficial learned behaviors; misaligned assumptions between components.	Restrict learning to high-level decisions or residual corrections; enforce hard safety constraints via WBC/MPC; residual RL on top of nominal controllers; structured interfaces and bounded corrections, validated in digital twins before deployment. [5], [10], [60], [61], [79], [80]

component is poorly calibrated, it may push the model-based core into rarely encountered states; conversely, if the model-based core is too conservative, it can negate the benefits of learning.

A common mitigation strategy is to *constrain the role of learning* to parts of the pipeline that are less safety-critical: for instance, using RL to generate base motion or high-level task commands while an MPC or WBC layer enforces joint limits, friction cones, and collision avoidance. [5], [10] Another strategy is residual learning, where the learned module only adds a bounded correction on top of a nominal model-based policy, making it easier to analyze and constrain its effect. [61], [71] Digital twins make it possible to test these hybrids under injected failures and parameter drifts before field deployment.

Overall, Table 26 highlights that model-based methods primarily rely on careful identification and conservative margins, learning-based methods rely on randomization and real-data fine-tuning, and hybrid methods attempt to combine both via structured interfaces and residual learning.

The choice of simulation and digital-twin stack also influences how easily results can be reproduced and compared across groups. In the next section The article synthesize the trends observed across modeling, tasks, control methods, coordination challenges, and tools into a concise summary before concluding with open research directions.

VIII. SUMMARY

This review has centered quadruped loco-manipulation around a three-way structure of *modeling*, *task taxonomy*, and *control methods*, with impedance/force control, simulation, and digital twins as cross-cutting ingredients. In contrast to prior surveys on mobile manipulation, legged locomotion, and whole-body control (compared in Table 1), we adopted a unified floating-base/contact model (Sec. III) and a six-axis loco-manipulation task taxonomy (Sec. IV, Table 4) as the common backbone for analyzing model-based, optimization-based, learning-based, and hybrid controllers.

On the **modeling** side, we formalized full floating-base rigid-body dynamics with contacts (Eqs. (2)–(8), (3)), and centroidal momentum dynamics (Eq. (6)), and showed how these appear in practice as five main abstractions: decoupled base+arm models, whole-body RBD, reduced-order/centroidal models, hybrid/complementarity contact formulations, and learned or residual dynamics (Figure 5, Table 2). These abstractions underpin the controllers surveyed later: whole-body inverse-dynamics stacks operate on the full RBD model, many MPC formulations work on centroidal or mixed models, and learning-based approaches often wrap or augment these with data-driven residuals.

On the **control** side (Sec. V), we contrasted four method families using the shared modeling backbone and task axes. Model-based inverse-dynamics whole-body control

TABLE 27. Loco-manipulation-specific sim-to-real difficulty by task regime (mapped to the six taxonomy axes in Section IV). The hardest-to-transfer regimes are those combining (i) contact-rich interaction at the end-effector, (ii) uneven/uncertain footholds, and (iii) latency/occlusion in perception.

Task regime (taxonomy-aligned)	Transfer difficulty	Why it is hard to transfer (gap mechanism)	Typical failure manifestation on hardware	Mitigations reported / suggested
Dynamic manipulation while walking (dynamic + prehensile/non-prehensile)	High	Arm-induced wrench coupling and fast contact switching stress both foot and hand contact models; prediction drift accumulates under model mismatch [28], [56].	Base drift, slip, missed grasps, solver overruns or unsafe transients during mode switches.	Disturbance-aware prediction; hierarchical decomposition; conservative feasibility margins; impedance shaping [55]–[57].
Uneven/compliant terrain with interaction (terrain uncertainty + contact-rich)	High	Foot micro-contact and friction/compliance mismatch dominate; contact timing and GRF distribution differ from simulation; small errors produce slip and force spikes [27], [65].	Unexpected slipping, contact loss, degraded force tracking at end-effector.	Contact margins/blending; passivity/robust WBC; domain randomization over friction/compliance [2], [65].
HRI/handover and force-limited interaction (strict safety + compliance)	High	Force/impedance must be correct at the end-effector while the base remains stable; safety constraints depend on accurate sensing/latency and contact models [20], [59].	Force overshoot, discomfort/unsafe impacts, saturation-induced instability.	Task-space impedance envelopes; explicit constraints; safety filters around learned policies [10], [20], [59].
Vision-conditioned object interaction (sensing axis: proprio+vision)	Medium–High	Photometric and geometric domain gaps (lighting, textures, occlusion) plus perception latency; interaction becomes partially observed [2], [58].	Grasp/approach failures, delayed reactions, policy collapse under occlusions.	Synthetic data + domain randomization; real demonstrations; latency/noise injection; hybrid fallback controllers [2], [81].
Quasi-static pick/place on flat ground (structured contact)	Low–Medium	Lower dynamic coupling; fewer rapid mode switches; contact conditions more repeatable.	Moderate tracking error due to inertial/friction mismatch.	Conventional calibration; conservative friction cones; log-replay digital twin tuning [101], [102].

(ID-WBC) uses the full floating-base equations and explicit contact constraints to coordinate base and arm motion, enforce joint/torque/friction limits, and realize prioritized task hierarchies (Eqs. (24), (25)–(28), Alg. 2). Representative designs and their practical trade-offs are summarized in Table 8. These stacks are strong for precise, constraint-aware loco-manipulation on moderately structured terrain but depend on accurate models and real-time QP solves.

Optimization-based methods, particularly whole-body and hierarchical model predictive control (MPC), extend this picture to finite horizons (Eqs. (30)–(31)), explicitly anticipating contact switches, load transfer, and base–arm coupling (Figure 8, Alg. 3). Whole-body MPC, disturbance-compensation MPC, hierarchical MPC, and data-driven MPC variants were compared in Tables 9 and 10. These approaches provide predictive coordination and principled handling of ground reaction forces but expose significant computational and modeling demands, especially for high-DoF quadrupeds with arms and rich perception.

Learning-based controllers (Sec. V-C)-including model-based RL, model-free RL, imitation learning, and data-driven MPC-use policies π_θ (Eqs. (32)–(34)) or learned dynamics models to absorb unmodeled contacts, terrain variation, actuator effects, and perception noise. Recent unified whole-body policies and vision-conditioned controllers directly synchronize legs and arm from high-dimensional observations. Table 12 compares generalization, adaptability, and data efficiency across learning families, while Table 13 outlines regime-dependent selection guidance. These methods can produce behaviors that are hard to hand-engineer, but they typically require large-scale simulated data, careful sim-to-real pipelines, and often rely on model-based safety layers.

Hybrid controllers (Sec. V-D) deliberately combine these strengths: model-based WBC or MPC provides feasibility and hard constraints (joint limits, friction cones, collision avoidance), while learning supplies residual corrections, high-level decisions, or cost shaping. Impedance and force-control layers (Sec. V-E) act as a safety and interaction ''glue'' across families, providing compliant behavior for human–robot collaboration and contact-rich tasks, and are frequently embedded as task-space objectives within QPs or as post-filters on learned commands. Emerging multi-task, force-aware, and data-centric paradigms that blend these ingredients are summarized in Table 18 and Sec. V-F.

We highlighted four recurring **whole-body coordination challenges**-tight coupling between locomotion and manipulation, stability during dynamic manipulation, foot–ground contact modeling and control, and momentum/balance regulation (Sec. VI, Figure 6, Table 22). ID-WBC handles coupling and contact consistency well when models are accurate; MPC excels at dynamic stability and momentum regulation when solver rates and state estimation are sufficient; learning-based approaches offer adaptability to unmodeled effects but need safety scaffolding; and hybrid methods distribute

these challenges across components best suited to them. These same challenges motivate the increasing reliance on high-fidelity simulation and digital-twin infrastructures.

On the **tools and platforms** side, we reviewed simulation environments and digital-twin frameworks for loco-manipulation (Sec. VII), comparing physics fidelity, manipulator support, RL integration, and digital-twin readiness (Table 24). We then analyzed sim-to-real gap patterns and mitigation strategies across control families (Sec. VII-D, Table 26), emphasizing the complementary roles of model identification, domain randomization, residual learning, safety wrappers, and hardware-in-the-loop twins. Finally, we surveyed reported quadruped+arm platforms (Table 23), highlighting typical mass and payload ratios, arm DoF choices, and the prevalence of Separate Systems vs. Combined Systems planning approaches.

Overall, the survey shows that inverse-dynamics whole-body control grounded in full rigid-body dynamics (Figure 5, Alg. 2) offers precise, constraint-aware execution; MPC-style optimal control (Figure 8, Alg. 3) provides predictive coordination under explicit limits; learning-based methods supply robustness to unmodeled dynamics and perception; and hybrid architectures seek to balance these strengths. Progress toward robust quadruped loco-manipulation in industrial and urban environments will hinge on compute-efficient hybrids with principled safety layers, tighter simulation–hardware alignment (Tables 24 and 26), and improved methods for momentum and contact management (Eqs. 6 and 29; Table 22), all evaluated against task-taxonomy-driven benchmarks that span the interaction, terrain, payload, coordination, and sensing axes introduced in this review.

IX. CONCLUSION AND FUTURE SCOPE

This review synthesized loco-manipulation control for quadruped robots with manipulators along a modeling-tasks-controls triad. On the modeling side (Section III), we showed how whole-body rigid-body dynamics, centroidal abstractions, hybrid/complementarity formulations, and learned residual models form a spectrum of approximations that are reused across the literature. On the task side (Section IV), we proposed a six-axis taxonomy-interaction type, dynamics, terrain, payload, coordination, and sensing-derived from a bottom-up coding of published experiments, and argued that these axes are approximately orthogonal from a control-design point of view. On the control side (Section V), we critically compared four families of methods (model-based WBC, optimization-based MPC/TO, learning-based policies, and hybrids) using these task axes and the whole-body coordination challenges in Section VI as evaluation lenses.

Three cross-cutting insights emerge from this analysis. First, *model choice is rarely neutral*: whole-body RBD with explicit contact constraints is effectively a prerequisite for high-end ID-WBC and whole-body MPC, whereas centroidal and decoupled base+arm models dominate when compute or software complexity must be constrained. Second, the six task axes reveal clear regime changes: increasing terrain difficulty, payload ratio, and dynamicity tends to push the field from purely model-based approaches toward MPC and hybrid or learning-based controllers, while moving from proprio-only to proprio+vision sensing shifts the emphasis from QP-structured WBC to perception-aware MPC and RL. Third, the coordination challenges in Section VI-tight base-arm coupling, disturbance rejection during dynamic manipulation, accurate foot-ground force regulation, and momentum control-systematically expose the limits of each control family: ID-WBC struggles with model mismatch, MPC with real-time computation and perception latency, and pure RL with safety and contact feasibility.

Model-based stacks remain the most mature option for safety-critical, prehensile tasks on moderately structured terrain. They offer interpretable task hierarchies, explicit torque and friction-cone limits, and well-understood tuning workflows, but require accurate inertial/contact models and careful gain and priority design. Optimization-based approaches, especially whole-body and hierarchical MPC, add explicit look-ahead over contact changes and arm-induced disturbances, which is essential for dynamic loco-manipulation with significant payloads. However, they are tightly coupled to solver capabilities and to the quality of state estimation and environment models. Learning-based methods, including unified whole-body policies and vision-conditioned controllers, provide robustness to unmodeled effects and perception noise, and have recently demonstrated impressive multi-task loco-manipulation behaviors, but they still depend on extensive data, careful sim-to-real pipelines, and often rely on model-based layers to guarantee basic safety. Hybrid controllers explicitly exploit this complementarity by delegating hard safety and feasibility constraints to model-based modules while using learning for residual dynamics, high-level decision-making, or cost shaping.

Future scope: taxonomy- and challenge-driven directions. Rather than listing generic open problems, we use the taxonomic axes in Section IV and the coordination challenges in Section VI to identify research threads that directly address gaps highlighted earlier (benchmark fragmentation, contact/latency mismatch, and deployability of hybrid stacks):

- *Task-axis-driven benchmarks and decision tools (taxonomy → benchmark design):* The six-axis taxonomy suggests a principled way to *parameterize* benchmarks: each benchmark instance is a ''task card'' defined by an axis tuple (interaction type, terrain uncertainty, payload ratio, sensing, contact richness, dynamics). A concrete direction is to publish an *axis-balanced* benchmark suite (e.g., factorially balanced or orthogonal-array subsets of the full grid) with open robot models, disturbance scripts, and logging protocols. Beyond success rate, protocols should report coordination metrics aligned with Table 22 (e.g., GRF/friction-cone margins, centroidal momentum regulation error,

and end-effector interaction-force envelopes), enabling quantitative "when-to-use-what" comparisons across WBC, MPC, RL, and hybrids. High-dynamic legged-manipulation demonstrations further motivate benchmark standardization in contact-rich regimes [104].

- *Coordination-aware hybrid architectures (coordination gaps → deployable hybrids):* The coupling and disturbance challenges in Section VI indicate that *industrial-grade* solutions will likely be hybrid: a safety-critical backbone enforces contact feasibility and limits, while learning handles perception-conditioned strategy and residual adaptation. Promising deployment-oriented patterns include: (i) *MPC/ID-WBC backbone + learned residuals* (e.g., residual dynamics/wrench prediction or cost shaping) to improve robustness without relaxing constraints [61]; (ii) *learned reference generation + QP/MPC safety filters*, where learning proposes footholds/end-effector references and optimization enforces friction cones, joint limits, and collision constraints; and (iii) *fallback-safe hierarchical stacks*, where a conservative controller guarantees stability and the learned layer is gated by runtime monitors (latency/freshness, contact feasibility, and force limits). These patterns align with industrial requirements for determinism, debuggability, and fail-safe behavior.
- *Force- and constraint-aware learning (learning gaps → contact realism):* Extending learning to the full task space requires objectives and representations that encode contact modes, safety margins, and feasibility costs, rather than purely kinematic tracking. Recent hierarchical RL results in constrained settings illustrate progress toward structured policies that respect environment constraints [76]. A key direction is coupling such policies with model-based contact predictors (for feasibility) and explicit force/impedance envelopes (for safety).
- *Human-centric loco-manipulation and intent-aware control (HRI gaps → comfort/safety metrics):* For handover and collaborative tasks, future systems should combine whole-body stability guarantees with models of human intent and comfort. This motivates ergonomics-aware optimization terms (manipulability, posture, comfort proxies) together with learned predictors of human motion/preferences, embedded inside hybrid stacks that explicitly trade off stability, task performance, and user comfort.
- *Digital-twin-driven design loops and failure analysis (sim gaps → reproducible stress tests):* Digital twins should be used not only for training but also for controlled failure analysis: log replay, parameter identification, injected latency, and rare-disturbance stress tests. A concrete direction is to release benchmark packages that bundle twin calibration tools, containerized baselines, and standardized evaluation to reduce configuration drift and improve reproducibility [105]. Such packages would allow systematic search over controller structures (task hierarchies, hybrid decompositions) and impedance parameters to maximize robustness along the coordination challenges in Table 22.

Overall, the review indicates that the most promising path forward is not a single "best" control paradigm but a carefully engineered combination of modeling abstractions, task-aware taxonomies, and hybrid control architectures. Future quadruped loco-manipulation systems are likely to rely on computationally lean cores that enforce contact feasibility and stability, wrapped by data-driven modules that provide adaptability across tasks, terrains, payloads, and human partners, and evaluated against standardized, taxonomy-grounded benchmarks in both simulation and digital-twin deployments.

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